DHCNet: Dynamic Hierarchical Collaborative Learning for Change Detection in High-Resolution Remote Sensing Images

Weilong Chen, Qing Ding, Fengyan Wang, Mingchang Wang, and Kaiyuan Sun

***Abstract*—High-resolution remote sensing (HRRS) image change detection (CD) can effectively monitor dynamic changes of the Earth's surface. In recent years, deep learning-based remote sensing image CD (RSCD) methods have developed rapidly. However, they still have shortcomings in spatio-temporal feature interaction of images, multiscale feature enhancement (MSFE), and multilevel information fusion. Therefore, this article proposes a dynamic hierarchical collaborative learning method (DHCNet) for accurate RSCD. First, a dynamic hierarchical collaborative interaction (DHCI) module is designed. It improves the richness of bi-temporal multigranularity features through layer-differentiated feature exchange and cross-temporal collaborative attention mechanism. Second, an MSFE module is constructed to enhance DHCNet's ability to represent the texture and semantic information of changed ground objects. Furthermore, a multilevel feature reconstruction (MLFR) module is proposed. It deeply fuses the enhanced multilevel features to reconstruct key change information and improve the model's CD performance. Experimental results on the LEVIR-CD, WHU-CD, CDD, Real-Wuhan, and S1GFloods datasets show that DHCNet's CD outperforms existing state-of-the-art methods, with $F1$-scores reaching 91.55%, 95.73%, 96.40%, 75.63%, and 96.36%, respectively. Comparative and ablation experiments confirm that DHCNet has significant advantages in CD of complex scenarios and fine-grained boundary representation. Our source codes are available at https://github.com/WBJIIKe/DHCNet.**

***Index Terms*—Change detection (CD), dynamic hierarchical collaborative interaction (DHCI), remote sensing images, transformer.**

I. Introduction

CHANGE detection (CD) technology analyzes, identifies, and monitors changes occurring to a target or scene over different periods. As a key area in the application of remote sensing technology, remote sensing image CD (RSCD) plays an important role in urban planning [1], agricultural monitoring [2], and environmental protection [3]. With the rapid development of satellite sensor technology, high-resolution remote sensing (HRRS) images have become increasingly abundant and are widely used in CD research. However, high-resolution images are rich in texture information and have complex spatial structures.

Traditional CD methods use algebraic calculations, image transformations, and postclassification comparisons [4], [5], [6]. They are inefficient and have low accuracy in complex scenarios. Therefore, there is an urgent need to develop efficient and accurate CD algorithms. The rapid development of deep learning technology has provided new ideas for CD tasks. Supervised CD methods based on deep learning have demonstrated excellent performance in very high-resolution bi-temporal images.

As a mainstream technology in deep learning, convolutional neural networks (CNNs) possess powerful feature extraction capabilities. Through convolution operations, they can capture ground objects' spatial structures and semantic features in remote sensing images [7], [8]. These features can fully express the change information between the complex bi-temporal images and are used to generate accurate RSCD results. Although CNN-based methods for RSCD [9], [10] have achieved significant success, they still face limitations. Specifically, CNN models are constrained by fixed receptive fields and a limited ability to interpret deeper-level information. These constraints prevent CNN models from effectively extracting large-scale spatial texture information hidden in remote sensing images and identifying salient regions in HRRS images. Consequently, this shortcoming restricts the algorithm's ability to distinguish specific complex changes in ground objects.

To address these limitations, researchers have introduced transformer [11], [12] into RSCD. Initially designed for natural language processing tasks, transformer has been adopted in computer vision research due to its exceptional ability to model long-range dependencies. In RSCD, transformer excels at modeling the long-distance contextual dependencies hidden in HRRS images. It captures global contextual information and processes remote correlated features, enabling more accurate capture of patterns in change evolution. Therefore, various transformer-based models have been widely applied to RSCD tasks [13], [14], [15]. Recently, some Mamba-based methods have shown good performance and have been quickly applied to CD [16], [17].

Although transformer-based models have demonstrated superiority in RSCD, there is still room for improvement.

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1) Traditional transformer models are good at capturing long-range dependencies and overall semantic information. However, their ability to extract and enhance local multiscale textures is limited, and they pay little attention to such textures.

2) The CD task requires comparing remote sensing images from different phases to identify the locations of changes. This process needs to consider the complex information associations and spatio-temporal cues among multitemporal images. However, current methods only focus on the internal relationships within single-phase images or simple interaction patterns of bi-temporal features. They neither fully consider the multilevel feature connections between bi-temporal images nor pay attention to the critical feature interactions.

3) Many existing transformer-based CD methods do not fully utilize the multilevel features generated in the encoding stage when restoring spatial dimensions during feature decoding. Instead, they merely enlarge feature map sizes through upsampling. This approach neglects the effective fusion of high-level semantic and spatial texture information from different feature levels. In addition, they lack mechanisms to reconstruct key change information and improve the precise identification of change boundaries, negatively impacting CD results.

To address these issues, we propose an innovative method named DHCNet in this article. It combines the advantages of CNNs and transformers while focusing on the spatio-temporal cues between bi-temporal images. The main contributions are as follows.

1) We propose the dynamic hierarchical collaborative interaction (DHCI) module. Dynamic hierarchical feature exchange and cross-temporal collaborative attention mechanism enable the interaction of bi-temporal features and effectively capture spatio-temporal correlation cues.

2) We construct a multiscale feature enhancement (MSFE) module to enhance the representational capability of multiscale texture and semantic information, addressing the issue of insufficient local feature extraction.

3) We design a multilevel feature reconstruction (MLFR) module. A cross-level feature fusion mechanism enables the effective reconstruction of features across different levels, significantly enhancing the refined identification capability of change boundaries.

4) Comparative and ablation experiments were conducted on the LEVIR-CD, WHU-CD, and CDD datasets. The generalization ability of the DHCNet was tested on the Real-Wuhan and S1GFloods datasets. The results show that DHCNet demonstrates significant advantages and exhibits leading robustness.

II. Related Work

A. CNN-Based CD Methods

In recent years, CNNs have become an indispensable technology in CD tasks. CNN-based CD models can be divided into single-stream and dual-stream models. In single-stream frameworks, the two temporal images are first fused and then input into the neural network to extract differential features. Peng et al. [18] connected the image pair as input and employed a multiside output fusion strategy to generate the final change map. Lin et al. [19] interpreted CD as a video understanding problem, constructing input image pairs into a pseudo-transitional video with rich temporal information. Wang et al. [20] proposed a hybrid affinity matrix that can more effectively explore the change patterns between the corresponding spectral vectors of spatial pixels. However, the single-stream structure prematurely introduces change information into semantic modeling, resulting in an insufficient representation of the land cover features in the individual original images, leading to relatively coarse CD results.

With the deepening of research, it has been found that the twin structure is more suitable for RSCD than the single-stream structure. The dual-stream structure is essentially a twin network architecture. Its key characteristic is to simultaneously train two subnetworks to learn the deep feature representations of the bi-temporal images, then fuse the features from bi-temporal images, and finally generate the CD map. Daudt et al. [21] proposed a twin neural network with weight-sharing encoders based on U-Net, improving the network's accuracy and inference speed. Chen et al. [22] introduced a dual-attention mechanism to capture long-range dependencies in the image, obtaining more discriminative feature representations. Zheng et al. [23] proposed a built-in high-frequency attention block, which enhanced high-frequency information for buildings and enabled better detection of the change edges of buildings. The dual-stream structure network can more accurately capture the relationships and similarities between features, leading to significantly better CD performance than single-stream networks [24], [25]. Although CNN-based methods have made significant progress in RSCD tasks, challenges remain, such as insufficient local feature extraction, limited global receptive fields, and poor adaptability to complex scenarios.

B. CNN-Transformer Hybrid CD Methods

Notably, in the evolutionary process of CD methods from CNN-based to transformer-based approaches, CNN-transformer hybrid methods have played a pivotal role, effectively addressing the inherent limitations of the two early single architectures. CNNs excel at extracting local spatial details and texture features, which is crucial for capturing fine-grained changes such as edges of newly constructed buildings, but they have obvious limitations in modeling long-range spatio-temporal correlations in large-scale remote sensing images; while early pure transformers, despite their outstanding performance in global dependency modeling, face issues of insufficient preservation of local features and high computational complexity.

By fusing the advantages of the two architectures, hybrid methods achieve the synergistic representation of local detail perception and global semantic correlation. Yuan et al. [26] proposed STransUNet, which can not only capture shallow detail features in the early stages but also model global context in high-level features. Liu et al. [2] proposed MSCANet, which employs a CNN-based feature extractor to capture hierarchical features and then designs a transformer-based MSCA to encode and aggregate contextual information. Li et al. [27] proposed MCTNet, which utilizes multiscale global and local

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information to enhance the robustness of CD in regions of different sizes. These hybrid methods not only alleviate the flaws of single architectures but also provide important technical insights for the evolution of CD methods toward the pure transformer paradigm.

C. Transformer-Based CD Methods

In recent years, the transformer structure has rapidly developed and updated, and its feature extraction and detail expression capabilities have been increasingly enhanced. Due to its strong capability in modeling long-range dependencies, researchers have extended the application of transformers [28] to computer vision, such as in vision transformer (ViT) [29], [30], Swin transformer [31], [32], and deformable transformer [33]. Currently, transformers have been shown to have considerable growth potential in CD [34], [35]. They can capture global dependencies in remote sensing images, possess a large receptive field, and are well-suited for detecting complex change patterns.

Transformer-based CD methods can be broadly categorized into two primary groups. In the first group, existing transformer architectures and their variants serve as either encoders or decoders for feature extraction and reconstruction. For example, BIT [36] employs a transformer encoder to model the contextual relationships among semantic tokens. Subsequently, it maps these features back to the pixel space using a transformer decoder. Bandara and Patel [37] combine a hierarchical transformer encoder with a multilayer perceptron (MLP) decoder to capture the multiscale and remote details required for effective CD. Zhang et al. [38] introduced an all-transformer network wherein both the encoder and decoder utilize Swin transformer blocks as their basic units, enabling the efficient extraction of global spatio-temporal information. ScratchFormer [39] applies a random sparse attention mechanism as its encoder to capture the inherent characteristics of the data and enhance the responsiveness to semantic changes. Moreover, Tang et al. [40] integrate deformable convolutions [41] and a deformable attention mechanism into both CNN and transformer branches, thus mitigating the adverse effects of fixed convolution kernels and regular patch segmentation. Pang et al. [42] designed a semantic feature-guided CD framework, utilizing the prior information of building semantic features to reduce the CD task's dependence on sample data effectively. Chen et al. [43] proposed the transformer-based object-level CD framework, which greatly alleviated the inherent limitations of pixel-level detection. Dong et al. [44] proposed a self-supervised weighted contrastive learning method, which has good generalization ability and performs excellently in CD of multiresolution synthetic aperture radar (SAR) images. ADDEDNet [45] incorporates a multiscale strip convolution structure to dynamically allocate attention weights, and embeds an adaptive attention mechanism in the decoder for enhanced feature expression. Xie et al. [46] proposed a FAST encoder based on window-based multihead self-attention and shifted window-based multihead self-attention to output bi-temporal multiscale features.

The second category embeds transformers and their variants as functional modules within the network architecture. By enhancing the transformer framework, Cao et al. [47] introduced a cross-scale feature interaction module that improves the feature interaction and fusion between scales. Similarly, TransUNetCD [48] incorporates a transformer module to capture global context and deep features, thereby reducing interference from pseudo-changes. Lin et al. [49] designed the token exchange-based difference evaluation module, which exchanged tokens of bi-temporal features and conducted difference feature modeling, making change differences more prominent. MSNet [50] incorporated a multiscale attention mechanism to extract local multiscale features from bi-temporal images. Wang [51] designed edge and global attention mechanisms, enabling the model to extract and reinforce features at different levels and facilitate effective multiscale feature fusion. Despite the advantages of transformer-based models in modeling long-range dependencies for RSCD tasks, current methods still encounter several challenges. These include balancing global semantics with local details, effectively establishing spatio-temporal associations across multilevel features, and optimizing the reconstruction process to integrate semantic information and spatial details from various feature hierarchies during decoding.

III. Methodology

A. Overview of the Proposed DHCNet

To overcome issues such as limited spatio-temporal feature interaction, insufficient MSFE, and inadequate multilevel information fusion that affect CD quality, we propose DHCNet in this article. Fig. 1 illustrates the basic architecture of DHCNet. Due to CD task complexity, the backbone network must exhibit robust feature extraction capabilities. Considering both model parameters and computational efficiency, the Swin transformer is selected as the primary backbone for feature extraction. The Swin transformer employs a window-based multihead self-attention mechanism and a shifted window partition strategy, thereby reducing the computational complexity while effectively modeling long-range dependencies. Traditional CNNs are limited by the locality of convolution operations and struggle to capture global semantic correlations. Swin transformer efficiently fuses global and local features through a hierarchical shifted window mechanism. The number of heads in different stages increases with feature dimension, allowing the model to focus on texture differences in shallow layers and capture semantic differences in deep layers. Learnable relative positional biases not only model the spatial relationships of pixels within windows but also adapt to inputs of different resolutions through log-spaced continuous positional encoding. This design makes the model more robust when handling registration errors. The proposed network utilizes the Swin transformer backbone to model global contextual relationships and extract fundamental features from bi-temporal images. The encoder consists of four stages, with the output feature map channels set at 96, 192, 384, and 768, respectively. The window size per stage is constantly 7×7. The number of layers per stage is 2, 2, 6, 2, respectively, and the number of heads per stage is 3, 6, 12, 24, respectively.

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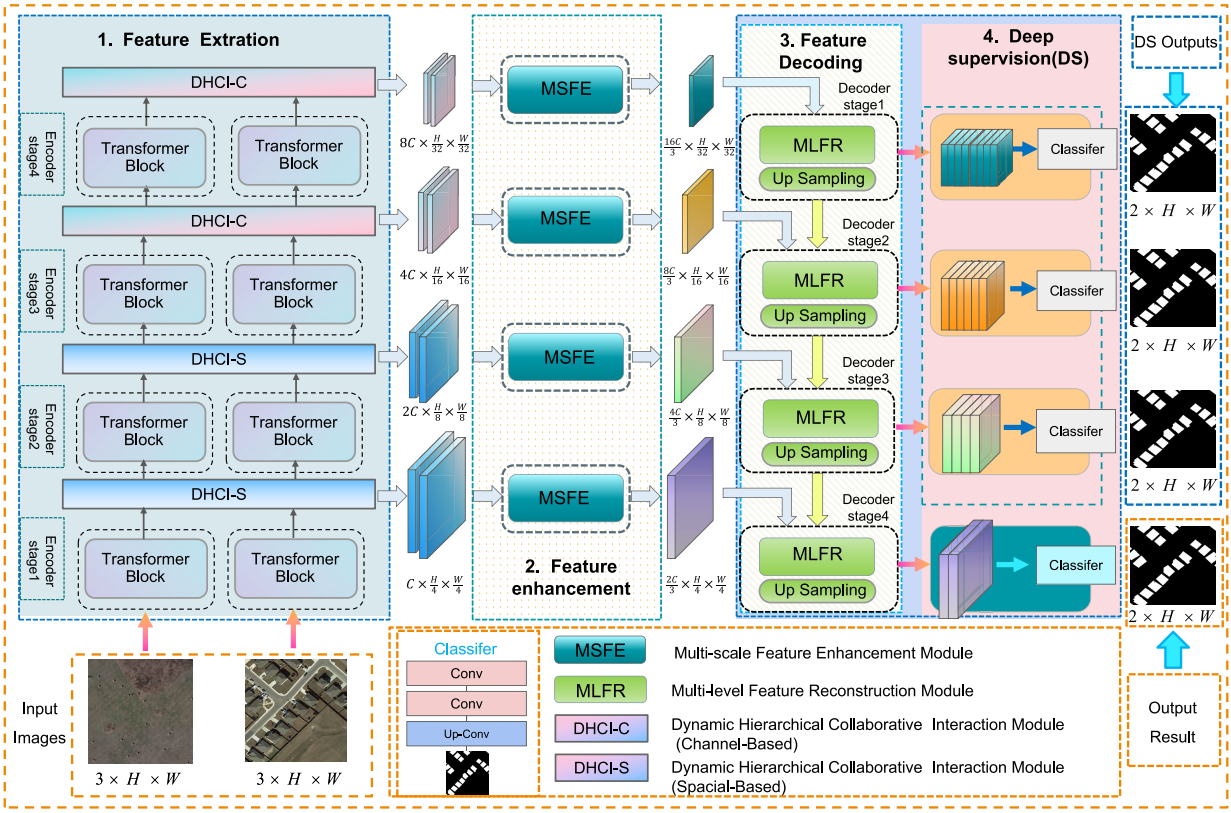


Fig. 1. Basic structure of the proposed DHCNet.

After the Siamese encoders complete feature extraction at each stage, it directly feeds the output bi-temporal features into the corresponding DHCI module for interaction. The features after interaction are further input into the next Swin transformer encoding stage, with a total of four stages. The DHCI module dynamically switches the interaction mode according to the feature characteristics of different encoder stages: in stages 1 and 2, the features are rich in spatial details but sparse in semantics, so the DHCI-S module is adopted for spatial-dimensional feature interaction; in stages 3 and 4, the features are concentrated in semantics but insufficient in spatial information, so the DHCI-C module is adopted for channel-dimensional feature interaction. The feature maps from the four encoder stages are then input into the corresponding MSFE module to enhance multiscale features. The resulting enhanced feature maps are subsequently fed into the decoder. In the decoder, the MLFR module, together with skip connections, fuses and reconstructs high-level semantic information and spatial texture details from different levels, enabling better identification of the types and properties of changes. Finally, each of the four decoder stages outputs classification results through classifiers, with deep supervision (DS) applied. The weights for DS across the various branches are discussed in Section V-E.

B. DHCI Module

Considering that single-temporal features have information limitations, we propose the DHCI module (see Fig. 2) to realize the hierarchical dynamic interaction of cross-temporal features. This module enhances the model's in-depth understanding of changed targets, improves the model's ability to capture dynamic contextual cues, and thus enhances CD performance.

The DHCI module consists of dynamic hierarchical feature exchange and collaborative attention feature fusion. The dynamic nature of the module is mainly reflected in its layer-specificity. Specifically, in the first two stages, the encoder outputs shallow features with sparse semantics but rich spatial details; thus, feature exchange is performed in the spatial dimension [52]. In the last two stages, the encoded features are semantically rich but lack detailed spatial information; therefore, feature exchange is conducted in the channel dimension instead. This article subdivides DHCI into two cases, namely, DHCI-S (spatial-based) and DHCI-C (channel-based). Both spatial and channel feature exchanges are based on the index divisibility rule: let the cross-temporal feature pair be $\{X_1, X_2\}$ (where $X_1, X_2 \in R^{h\times w\times c}$). Let $P \in [0, 1]$ (exchange ratio); when $P = 0$, no exchange is performed; when $P \neq 0$, $1/P \in N^+$. Let the spatial position index of features X_1 and X_2 be K (where $K \in \{1, 2, \ldots, M\}$) or the channel index be J (where $J \in \{1, 2, \ldots, N\}$). When K or J is divisible by the $1/P$, the values of cross-temporal features X_1 and X_2 are exchanged accordingly. Features with nondivisible indices remain unchanged to maintain local continuity and semantic integrity; features with divisible indices are forced to interact to achieve cross-temporal information flow. Salient changes in remote sensing images typically occupy a small fraction of the total image area. Setting a moderate exchange ratio (e.g., 1/4)

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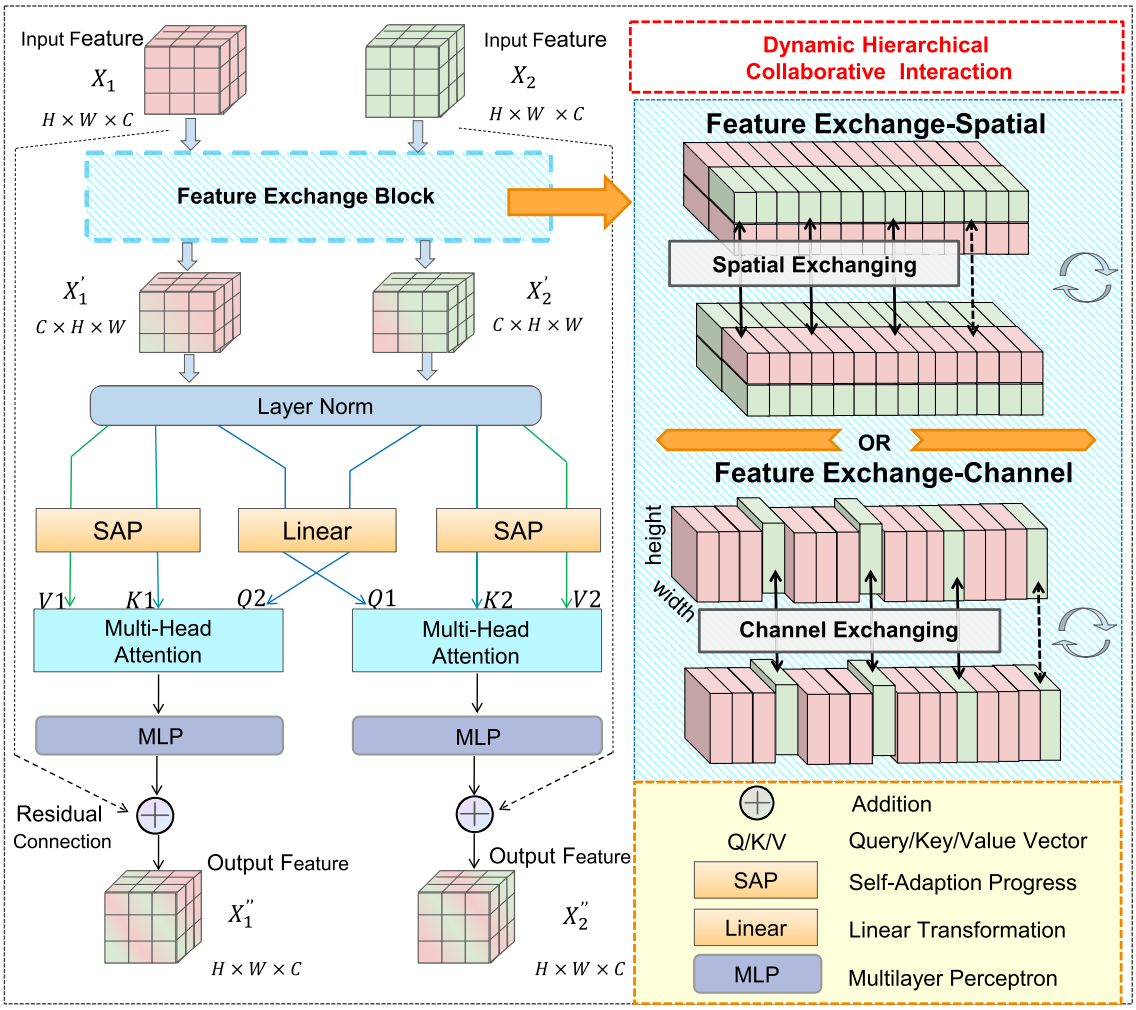


Fig. 2. Illustration of the DHCI module.

aligns with the sparsity assumption of information distribution in CD tasks, achieving a theoretical balance between preserving original feature integrity and facilitating cross-temporal feature interaction. A larger P leads to denser index screening, which may blur original features due to excessive exchange; a smaller P results in sparser feature exchange, which tends to cause insufficient information flow. In ablation experiments, the selection of P is discussed in depth. The postexchange features X_1' and X_2' are mapped to query vectors $\{Q_1, Q_2\}$, key vectors $\{K_1, K_2\}$, and value vectors $\{V_1, V_2\}$, with the process as follows:

$$X_n' = LayerNorm\left(X_n'\right) \tag{1}$$

$$Q_n = Reshape_2\left(Linear\left(Reshape_1\left(X_n'\right)\right)\right) \tag{2}$$

$$X_{tokv}' = Reshape_1\left(Conv_{1\times1}\left(Avgpool\left(X_n'\right)\right)\right) \tag{3}$$

$$K_nV_n = Reshape_2\left(LayerNorm\left(X_{tokv}'\right)\right) \tag{4}$$

$$K_n = K_nV_n\left[0\right], V_n = K_nV_n\left[1\right],\ \mathrm{n} \in \{1,2\} \tag{5}$$

where $Linear(\cdot)$ denotes the linear layer. $Reshape_1(\cdot)$ and $Reshape_2(\cdot)$ are used to realize the mutual conversion between tensors and vector sequences. $Conv_{1\times1}(\cdot)$ denotes the 1×1 convolution layer, and $Avgpool(\cdot)$ denotes adaptive average pooling. $LayerNorm(\cdot)$ denotes layer normalization. X_{tokv}' is the intermediate value in the calculation process.

Furthermore, the DHCI module generates key-value pairs through the combination of average pooling and convolution [53]. The pooling operation reduces the spatial dimensions of $\{K_n, V_n\}$, $n \in \{1, 2\}$ and lowers computational complexity. Subsequent convolutional operation further refines high-level semantic correlations from the pooled features. In the attention mechanism, the key vectors are transposed first, then undergo matrix multiplication with the query vectors. Furthermore, the dot product results are divided by the square root of the key vector dimension d to stabilize gradients. Next, the Softmax function is applied to the scaled dot product results, converting them into weights. These weights are then multiplied by the value vectors at corresponding positions to highlight important information in features. For the RSCD task, the DHCI module adopts a cross-temporal adaptive design for the attention mechanism. It swaps the query vectors Q_1 and Q_2 to calculate cross-temporal similarity, thereby mining the difference information of changed ground objects between different phases. Cross-temporal collaborative attention can further promote feature interaction between different phases and enhance the model's focus on changed targets. Finally, we

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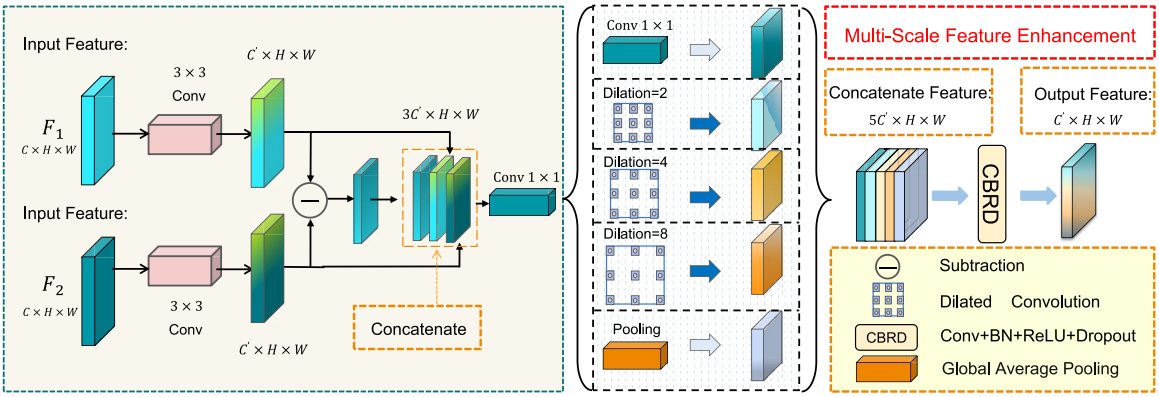


Fig. 3. Architecture of the MSFE module.

use residual connections to obtain the output features $\{X_1'', X_2''\}$, which improves information transmission and prevents gradient vanishing. The specific process is as follows:

$$MSA_j = Concat\left(head_1^j, \ldots, head_n^j\right) W^O \; j \in \{1, 2\} \tag{6}$$

$$head_m^1 = Attention\left(Q_2 W_m^Q, K_1 W_m^K, V_1 W_m^V\right) \tag{7}$$

$$head_m^2 = Attention\left(Q_1 W_m^Q, K_2 W_m^K, V_2 W_m^V\right) \tag{8}$$

$$Attention\left(Q, K, V\right) = softmax\left(\frac{QK^T}{\sqrt{d}}\right) V \tag{9}$$

$$X_1'' = Reshape\left(MLP\left(MSA_1\right)\right) + X_1 \tag{10}$$

$$X_2'' = Reshape\left(MLP\left(MSA_2\right)\right) + X_2 \tag{11}$$

where W^O, W_m^Q, W_m^K, and W_m^V are the learnable parameter matrixes. $Concat(\cdot)$, $Attention(\cdot)$, and $softmax(\cdot)$ are concatenation, self-attention, and softmax functions. $MLP(\cdot)$ is the multilayer perception. MSA_j denotes the multihead attention of two phases. n denotes the number of multiheads. $Reshape(\cdot)$ serves to align the tensor. d denotes the dimension of the key vector.

C. Msfe Module

After passing through the transformer encoder and interaction module, the resulting features exhibit rich global dependency relationships, effectively capturing the semantic connections between features at different spatial locations. However, they neglect to extract and enhance multiscale information. To address this issue, we propose the MSFE module. It can effectively enhance multiscale texture and semantic information while preserving global dependency relationships, thereby making the final feature representation more abundant and discriminative. The structure of the MSFE module is shown in Fig. 3. Assuming the input data are two bitemporal feature maps $F_1 \in R^{c \times h \times w}$ and $F_2 \in R^{c \times h \times w}$, the process begins with two independent 3×3 convolutional layers to finely capture local detail information, generating $F_1^1 \in R^{c' \times h \times w}$ and $F_2^1 \in R^{c' \times h \times w}$. A pixel-wise difference operation is then applied to compute the feature discrepancy between the two images, resulting in a difference feature map $F_{diff} \in R^{c' \times h \times w}$. Subsequently, $F_{diff} \in R^{c' \times h \times w}$, $F_1^1 \in R^{c' \times h \times w}$ and $F_2^1 \in R^{c' \times h \times w}$ are concatenated to form a new feature map $F_{conc} \in R^{3c' \times h \times w}$. This step aims to enhance the feature difference information without losing information. Then, a 1×1 convolution is applied to obtain $F_{1\times1} \in R^{c' \times h \times w}$, which can be used to integrate channel information and reduce the computational cost of subsequent operations. The atrous spatial pyramid pooling (ASPP) can expand the receptive field. By using multiple dilated convolutions in parallel and then fusing their results, it can effectively enhance the multiscale texture and semantic features. Therefore, we adjust the dilation rates of ASPP to adapt to our method. Finally, the ASPP module generates $F_{msfe} \in R^{c' \times h \times w}$, enhancing multiscale information. The above process can be formulated as

$$F_1^1 = Conv_{3\times3}\left(F_1\right), F_2^1 = Conv_{3\times3}\left(F_2\right) \tag{12}$$

$$F_{diff} = \left|F_1^1 - F_2^1\right| \tag{13}$$

$$F_{conc} = Concat\left(F_1^1, F_2^1, F_{diff}\right) \tag{14}$$

$$F_{1\times1} = Conv_{1\times1}\left(F_{conc}\right) \tag{15}$$

$$F_{mid}^1 = Conv_{3\times3}^{(r)}\left(F_{1\times1}\right) \tag{16}$$

$$F_{mid}^2 = Up\left(GAP\left(F_{1\times1}\right)\right) \tag{17}$$

$$F_{mid}^3 = Conv_{1\times1}\left(F_{1\times1}\right) \tag{18}$$

$$F_{msfe} = Conv_{1\times1}\left(Concat\left(F_{mid}^1, F_{mid}^2, F_{mid}^3\right)\right) \tag{19}$$

where $Conv_{3\times3}(\cdot)$, $\|$, $Concat(\cdot)$, $Conv_{1\times1}(\cdot)$, $Conv_{3\times3}^{(r)}$, $GAP(\cdot)$, and $Up(\cdot)$ denote 3×3 convolution, the absolute value, channel-wise concatenation, 1×1 convolution layer, 3×3 dilated convolution, global average pooling, and upsampling, respectively. We set the dilation rate $r = 2, 4, 8$. F_{mid}^i is an intermediate value calculated for subsequent use.

D. MLFR Module

To restore the high-level semantic and spatial texture information contained in features at different levels, we designed the MLFR module (see Fig. 4). First, obtain F_{msfe} from the MSFE module and input it into a parallel branch. In branch 1, the feature map F_{msfe} is processed through a max pooling layer, an average pooling layer, and an MLP, and then subjected to pixel-wise addition. These operations can distill each extracted channel's average and prominent feature information, facilitating the learning of complex relationships among channels. Then, the weight Q^1 of each channel is obtained through the Sigmoid function, which is used to

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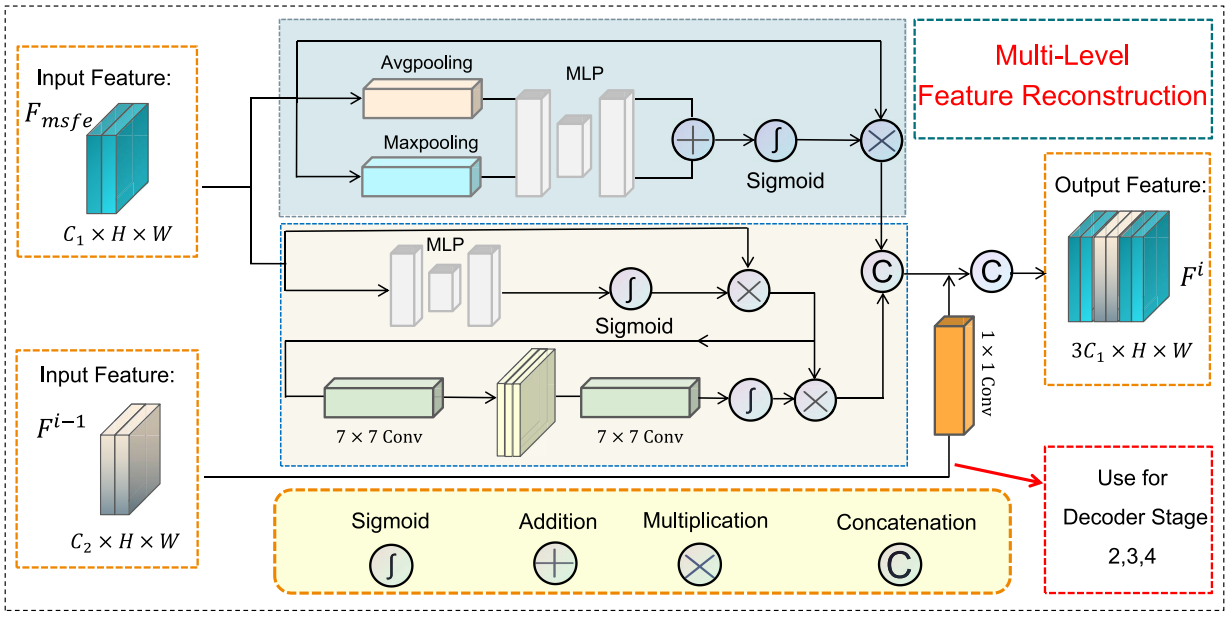


Fig. 4. Illustration of our MLFR module.

weight the original feature map. Through branch 1, important channel information can be highlighted, and features meaningful for RSCD can be extracted. In branch 2 [54], after F_{msfe} passes through an MLP and a Sigmoid function, it is multiplied pixel by pixel with the original feature map to obtain F_m. This step enables the integration of information and the elimination of useless interfering noise. Subsequently, benefiting from the noise purification process, after passing through two convolutional layers and a Sigmoid function, Q^2 is derived, which makes the extracted spatial features more precise. Finally, branch 1 and branch 2 are concatenated to obtain F_{mlfr}, enabling the acquisition of rich semantic and spatial texture information. Notably, at each decoder stage $i = 2, 3, 4$, F_{mlfr} is further concatenated with F^{i-1} to fuse semantic and spatial information across different scales. Herein, i denotes the index of the current decoder stage, and F^{i-1} refers to the output feature of the previous decoder stage relative to the current one. For instance, when the current decoder stage is 2, F^{i-1} represents the output feature from the first stage of the decoder. The above process can be expressed by the following equations:

$$Q^1 = \sigma\left(MLP\left(Maxpool\left(F_{msfe}\right)\right) + MLP\left(Avepool\left(F_{msfe}\right)\right)\right) \tag{20}$$

$$F_m = \sigma\left(MLP\left(F_{msfe}\right)\right) \times F_{msfe} \tag{21}$$

$$Q^2 = \sigma\left(Conv_{7\times7}\left(Conv_{7\times7}\left(F_m\right)\right)\right) \tag{22}$$

$$F^{i-1} = Conv_{1\times1}\left(Up\left(F^{i-1}_{mlfr}\right)\right) \tag{23}$$

$$F^{i}_{mlfr} = Concat\left(Q^1 F_{msfe}, Q^2 F_m, F^{i-1}\right) \tag{24}$$

where $MLP(\cdot)$ denotes the MLP function, $\sigma(\cdot)$ implies the Sigmoid function, $Conv_{7\times7}(\cdot)$ specifies the 7×7 convolution, $Conv_{1\times1}(\cdot)$ specifies the 1×1 convolution, and $Maxpool(\cdot)$ and $Avgpool(\cdot)$ mean the max and average pooling layer, respectively. $Up(\cdot)$ expresses the upsampling and $Concat(\cdot)$ indicates the concatenation operation at the channel-wise.

E. Loss Functions

In RSCD, class imbalance—where the number of changed pixels is much lower than that of unchanged pixels—represents a common issue. The binary cross-entropy (BCE) loss function helps alleviate the impact of this imbalance. Its computation is as follows:

$$L_{BCE}(X, Y) = -\left(Y log\left(X\right) + \left(1 - Y\right) log\left(1 - X\right)\right) \tag{25}$$

where Y is the ground truth and X is the predicted change map.

In RSCD, accurately delineating the boundaries of change regions is an indispensable step. The Dice loss function enhances the clarity and accuracy of the contours of change areas by comprehensively considering the overlap between the predicted and ground truth regions. It effectively suppresses undesirable phenomena such as blurred boundaries and over-prediction. The computation is as follows:

$$L_{Dice}(X, Y) = 1 - \frac{2XY + \gamma}{X^2 + Y^2 + \gamma} \tag{26}$$

where the γ can avoid the denominator from being 0. Ultimately, in order to mitigate the impact of sample imbalance and unclear boundaries, we combine the two aforementioned loss functions to construct our hybrid loss function, which can be expressed as

$$L = L_{BCE}(X, Y) + L_{Dice}(X, Y). \tag{27}$$

IV. Datasets and Experimental Setup

A. Datasets Introduction

To ensure the generalizability of DHCNet, we use five CD datasets selected for performance evaluation, as shown

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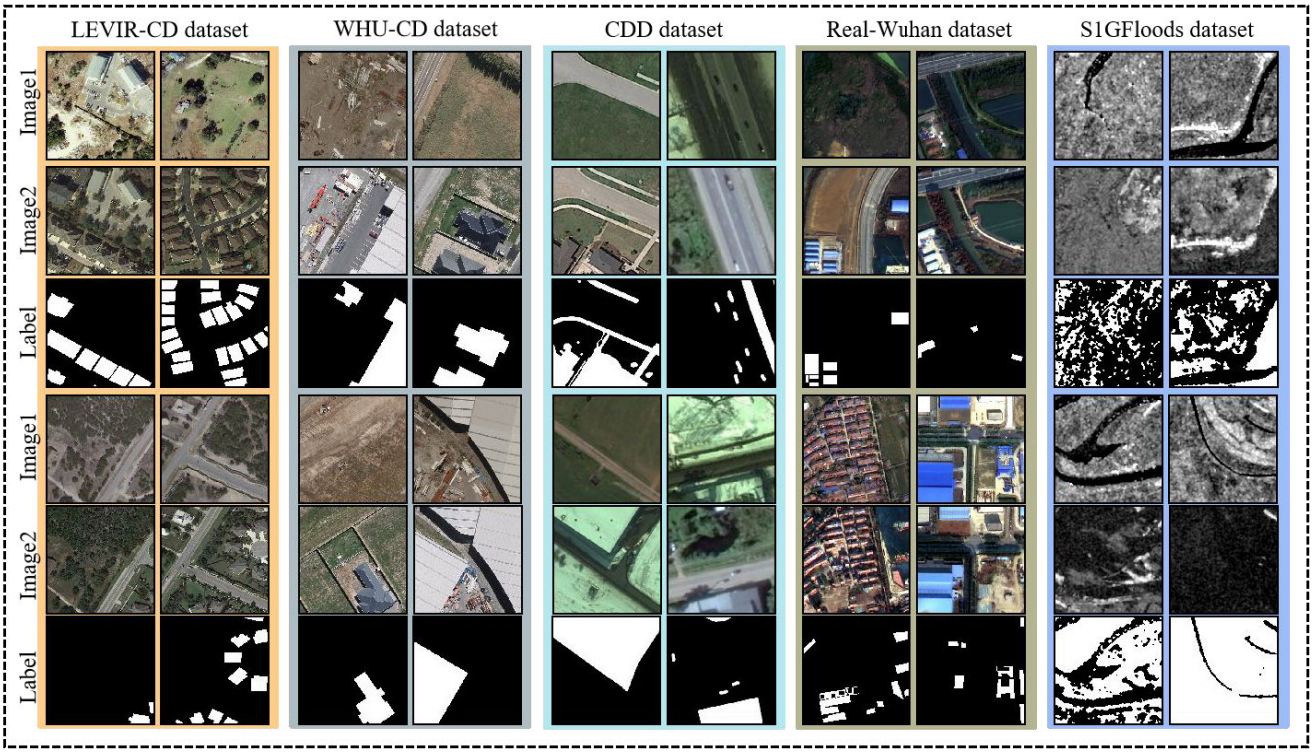


Fig. 5. Examples of LEVIR-CD, WHU-CD, CDD, Real-Wuhan, and S1GFloods datasets.

TABLE I
INTRODUCTION TO TRAIN/VAL/TEST DATASETS

Dataset	Size (pixel)	Number of Pixels Unchanged	Changed	Number of Images Pairs Train	validation	Test
LEVIR-CD	256×256	637,028,937	30,913,975	7,120	1,024	2,048
WHU-CD	256×256	466,085,729	21,108,895	5,974	744	733
CDD	256×256	914,376,178	122,427,529	10,000	3,000	3,000
Real-Wuhan	256×256	148,321,784	6,343,176	1,416	472	472
S1GFloods	256×256	237,710,465	113,562,495	3,751	536	1,073

in Fig. 5. These include the LEVIR-CD dataset [55], the CDD dataset [56], the WHU-CD dataset [57], the Real-Wuhan dataset, and the S1GFloods dataset [58]. The training, validation, and test sets are partitioned according to the scheme presented in Table I.

1) The LEVIR-CD dataset consists of 637 HRRS image pairs, each with a spatial size of 1024 × 1024 pixels and a ground sampling resolution of 0.5 m. The change objects include villas, apartments, townhouses, garages, warehouses, and parking lots. The original dataset is divided into 445 training pairs, 128 testing pairs, and 64 validation pairs. The image pairs are cropped into 256 × 256 pixel patches, resulting in 7120 training pairs, 2048 testing pairs, and 1024 validation pairs.

2) The CDD dataset contains 11 pairs of bi-temporal images with seasonal variations, with spatial resolutions ranging from 0.03 to 1 m. The 11 image pairs are cropped into 16 000 image patches of 256 × 256 pixels to facilitate training. Among them, 10 000 patches are used for training, 3000 for testing, and 3000 for validation. In the CDD dataset, the task is to detect changes caused by objects such as buildings, vehicles, and roads while ignoring changes induced by seasonal variations.

3) The WHU-CD dataset consists of a single pair of high-resolution images with a spatial resolution of 0.2 m and a spatial size of 32 507 × 15 354 pixels. Due to large-scale reconstruction following an earthquake, buildings represent the primary change objects in this dataset. The image pair is cropped into nonoverlapping patches of size 256 × 256 pixels. From these patches, 5974 pairs are selected for training, 744 for testing, and 743 for validation.

4) The Real-Wuhan dataset is constructed using Gaofen-2 satellite images covering Wuhan's main urban area in 2016 and 2021, focusing on high-resolution CD in complex urban scenarios. It primarily captures changes in typical urban features such as buildings, reflecting the dynamic surface evolution of Wuhan's urban core over five years. These bi-temporal image pairs are cropped into nonoverlapping 256 × 256 pixel patches. 1416 pairs of samples are randomly selected for training, 472 for validation, and 472 for testing. During the division process, we adopted a strict sample deduplication mechanism to ensure no sample overlap between the training set, validation set, and test set, thus guaranteeing the scientificity of data division and the reliability of experimental results.

5) S1GFloods is the world's first open-source large-scale flood detection dataset, constructed using Sentinel-1 SAR high-resolution imagery pairs covering 42 global flood events from 2016 to 2022. It comprehensively captures typical flood scenarios such as rivers, lakes, vegetated areas, and urban-rural built-up zones, thus truly reflecting flood dynamics

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across diverse land surfaces. To facilitate model training, the bi-temporal image pairs are cropped into nonoverlapping 256×256 pixel patches. The dataset is partitioned into 3751 training pairs, 1073 testing pairs, and 536 validation pairs.

B. Implementation Details and Reproducibility

All experiments in this study are conducted on a GeForce RTX 4090 using the PyTorch framework. All experiments adopt a unified random seed setting to ensure that all comparative methods, including DHCNet, can stably converge and achieve optimal accuracy. The comparison of experimental results exhibits fairness, reliability, and reproducibility. The AdamW optimizer is employed, with a weight decay of 0.01 and β values set to (0.9, 0.999). The initial learning rate is set to 0.0001 and decays linearly over time. The batch size is set to 16, and the maximum number of training epochs is 150. To guarantee the fairness of comparative experiments, all methods adopt the same data augmentation pipeline, including random horizontal flip, random crop, random angle rotation, color enhancement, adding salt-and-pepper noise, and Mosaic concatenation.

To ensure the reproducibility of the experimental results in this article, the experimental random seed is strictly fixed at 123 to eliminate the impact of randomness on the results. The GitHub repository contains the experimental code, the model weight files trained on the five benchmark datasets, as well as complete data preprocessing scripts and a custom PyTorch data loader. The datasets LEVIR-CD, CDD, WHU-CD, and S1GFloods used in this study are publicly available, widely adopted, and used for academic research in full compliance with their original license agreements. The Real-Wuhan dataset is self-constructed in this article, which is mainly used to verify the performance of DHCNet in real-world scenarios. This dataset will be made available upon request and is exclusively for academic research purposes.

C. Evaluation Metrics

To verify the performance of the proposed method, we use the following metrics for quantitative evaluation: $F1$-score ($F1$), precision (Pre), recall (Rec), overall accuracy (OA), Kappa coefficient (Kappa), and intersection over the union (IoU). Their definitions are as follows:

$$F1 = \frac{2}{Pre^{-1} + Rec^{-1}} \tag{28}$$

$$Pre = \frac{TP}{TP + FP} \tag{29}$$

$$Rec = \frac{TP}{TP + FN} \tag{30}$$

$$OA = \frac{TP + TN}{TP + TN + FN + FP} \tag{31}$$

$$Kappa = \frac{OA - Q}{1 - Q} \tag{32}$$

$$IoU = \frac{TP}{TP + FN + FP} \tag{33}$$

$$Q = \frac{(TP + FN) \times (TP + FP)}{(TP + TN + FP + FN)^2} + \frac{(TN + FP) \times (TN + FN)}{(TP + TN + FP + FN)^2} \tag{34}$$

where TP, TN, FP, and FN represent true positive, true negative, false positive, and false negative pixels, respectively. Q is an intermediate variable used for calculating kappa.

D. Comparative Methods

To demonstrate the superior performance of our proposed method, we compared DHCNet with representative CD approaches. We have reimplemented these methods and adopted the same data augmentation strategies as ours. Specific details are as follows.

1) FC-EF [21] has a U-shaped structure and uses concatenated images as the inputs. In addition, FC-EF incorporates several skip connection layers into the network.

2) FC-Siam-conc [21] adopts a Siamese U-Net architecture. Its encoder processes bi-temporal images simultaneously to generate multilevel features fused via concatenation.

3) FC-Siam-diff [21] has a Siamese U-shaped structure. The difference is that FC-Siam-diff uses an absolute difference operation to merge the bi-temporal features.

4) SNUNet [9] is a densely connected Siamese network. An ensemble channel attention module is applied for DS. Through ECAM, the most representative features at different semantic levels can be extracted.

5) BIT [36] is a bi-temporal image transformer network designed to capture and effectively model cross-spatial and temporal contextual information between bi-temporal images.

6) TinyCD [59] is a lightweight CD model using a novel spatio-temporal feature mixing strategy to fuse Siamese backbone embeddings.

7) ChangeFormer [37] incorporates the transformer architecture and accurately analyzes and detects changes between images from different time points by the self-attention mechanism.

8) IFNet [55] is a high-resolution RSCD method that uses a two-stream fully convolutional network for feature extraction and DS for CD.

9) DTCDSCN [60] performs building CD (pixel-level) and building extraction (object-level, semantic segmentation) simultaneously. The two tasks share the same feature extraction layer.

10) MAMI-CD [61] is a multilevel spatial attention network that can extract and transmit spatial attention features and strengthen target correlation through channel-based feature interaction.

11) DA-IDANet [62] reduces the interference of pseudo-changes by mapping the features of two images into a shared deep feature space for alignment. In addition, it designs the IDAM module to enhance the representation of features effectively.

12) ChangeMamba [63] is the first work to explore the Mamba architecture in the field of RSCD. Adopting the advanced vision Mamba as its encoder, it comprehensively captures global spatial contextual information.

V. Results and Discussion

A. Comparative Experiments on the LEVIR-CD Dataset

Table II presents the quantitative evaluation results of different methods on the LEVIR-CD dataset. Our method

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TABLE II
QUANTITATIVE COMPARISON RESULTS ON THE LEVIR-CD DATASET

Method	Pre(%)	Rec(%)	IoU(%)	F1(%)	Kappa(%)	OA(%)
FC-EF	84.59	76.19	66.91	80.18	79.17	98.08
FC-Siam-diff	86.82	78.08	69.80	82.21	81.31	98.27
FC-Siam-conc	88.50	81.67	73.84	84.95	84.18	98.52
SNUNet	94.12	87.00	82.52	90.42	89.93	99.06
IFNet	94.85	87.37	83.41	90.96	90.49	99.12
DTCDSCN	94.01	86.84	82.29	90.28	89.78	99.04
BIT	92.45	**88.98**	82.96	90.68	90.19	99.06
ChangeFormer	93.97	87.29	82.66	90.51	90.02	99.07
tinyCD	93.64	87.34	82.45	90.38	89.88	99.05
MAMI-CD	94.25	88.22	83.72	91.14	90.68	99.12
ChangeMamba	**94.98**	86.16	82.41	90.36	89.87	99.06
Ours	94.72	88.59	**84.42**	**91.55**	**91.12**	**99.16**

TABLE III
QUANTITATIVE COMPARISON RESULTS ON THE WHU-CD DATASET

Method	Pre(%)	Rec(%)	IoU(%)	F1(%)	Kappa(%)	OA(%)
FC_EF	68.41	21.04	19.18	32.19	30.82	96.26
FC-Siam-diff	73.54	55.69	46.39	63.38	62.00	97.29
FC-Siam-conc	70.15	45.89	38.39	55.49	53.95	96.90
SNUNet	93.18	80.05	75.62	86.12	85.56	98.91
IFNet	96.62	92.42	89.53	94.48	94.24	99.54
DTCDSCN	95.38	90.45	86.65	92.85	92.54	99.41
BIT	**97.20**	90.31	88.02	93.63	93.36	99.48
ChangeFormer	97.16	92.01	89.60	94.51	94.28	99.55
tinyCD	95.07	92.24	88.03	93.64	93.36	99.47
MAMI-CD	95.54	88.36	84.86	91.81	91.46	99.34
DA-IDANet	96.32	88.59	85.68	92.29	91.97	99.38
ChangeMamba	97.18	93.14	91.08	95.33	95.13	99.62
Ours	**97.20**	**94.30**	**91.80**	**95.73**	**95.54**	**99.65**

demonstrates superior performance to other methods on the LEVIR-CD dataset through a comprehensive evaluation of multiple indicators.

Experimental comparisons on the LEVIR-CD dataset reveal that DHCNet demonstrates superior overall performance. In terms of $F1$-scores, DHCNet outperforms ChangeMamba by 1.19%, DA-IDANet by 2.05%, MAMI-CD by 0.41%, tinyCD by 1.17%, BIT by 0.87%, DTCDSCN by 1.27%, SNUNet by 1.13%, FC-Siam-conc by 6.6%, FC-Siam-diff by 9.34%, FC-EF by 11.37%, ChangeFormer by 1.04%, and IFNet by 0.59%. These results indicate that DHCNet effectively balances Pre and Rec. In addition, DHCNet achieves the highest values in Kappa (91.12%), IoU (84.42%), and OA (99.16%), further validating the excellence of the proposed method in detection consistency and OA. Although DHCNet lags behind BIT by 0.39% and ChangeMamba by 0.26% in Rec and Pre, its multidimensional leadership in metrics such as IoU, $F1$, Kappa, and OA. These results fully demonstrate that DHCNet can more effectively balance various performance indicators and achieve optimal overall detection results on the LEVIR-CD dataset.

B. Comparative Experiments on the WHU-CD Dataset

We also conducted experiments on the WHU-CD dataset to validate the model's reliability, which features a smaller proportion of changed pixels and more complex scenarios of changed buildings. The quantitative results of different methods are presented in Table III.

In the quantitative comparison on the WHU-CD dataset, DHCNet outperforms all compared methods across all metrics. Specifically, with a Rec of 94.30%, it surpasses IFNet (92.42%), ChangeMamba (93.14%), and other methods, demonstrating a stronger ability to capture fundamental changes effectively. The $F1$-score of DHCNet reaches 95.73%, significantly better than IFNet (94.48%), DTCDSCN (92.85%), and others, which fully balances Pre and Rec. DHCNet's IoU of 91.80% indicates more accurate localization of changed regions. These results highlight that the proposed method exhibits strong scene generalization ability in urban building CD tasks. The leading values of Kappa (95.54%) and OA (99.65%) further demonstrate DHCNet's excellence in detection consistency and OA. Its core advantage lies in the innovative design of the interactive network architecture, which achieves the optimal balance between Pre and Rec in CD. The comprehensive leadership in comprehensive evaluation metrics ($F1$, OA, and Kappa) fully validates the effectiveness of the proposed method.

TABLE IV
QUANTITATIVE COMPARISON RESULTS ON THE CDD DATASET

Method	Pre(%)	Rec(%)	IoU(%)	F1(%)	Kappa(%)	OA(%)
FC-EF	80.31	38.62	35.28	52.16	48.19	91.64
FC-Siam-diff	86.28	50.88	47.07	64.01	60.56	93.25
FC-Siam-conc	82.23	52.77	47.37	64.29	60.66	93.08
SNUNet	91.52	82.30	76.46	86.66	84.98	97.01
IFNet	**98.05**	92.02	90.37	94.94	94.29	98.84
DTCDSCN	96.77	88.95	86.39	92.70	91.77	98.35
BIT	96.52	92.72	89.72	94.58	93.87	98.75
ChangeFormer	97.11	93.72	91.18	95.38	94.78	98.93
tinyCD	95.34	89.24	85.51	92.19	91.18	98.20
MAMI-CD	94.26	85.60	81.36	89.72	88.42	97.69
DA-IDANet	97.20	91.11	88.78	94.06	93.29	98.64
ChangeMamba	97.45	94.16	91.90	95.78	95.22	99.02
Ours	97.67	**95.16**	**93.04**	**96.40**	**95.92**	**99.16**

C. Comparative Experiments on the CDD Dataset

Besides building changes, we chose the CDD dataset, which includes various land cover changes. We evaluated how seasonal changes affect the method's detection results to check its robustness. The experimental results are shown in Table IV.

The proposed DHCNet outperforms the state-of-the-art in all metrics except Pre, showing significant advantages. Specifically, DHCNet leads by 0.62% in $F1$ and 1.14% in IoU. In terms of Rec, the DHCNet achieves 95.16%, surpassing IFNet (92.02%), BIT (92.72%), ChangeFormer (93.72%), and other methods. For the IoU metric, DHCNet reaches 93.04%, significantly leading over ChangeFormer (91.18%), IFNet (90.37%), and others, demonstrating stronger feature fusion ability and CD accuracy. Although the Pre (97.67%) of the DHCNet is slightly lower than IFNet's Pre (98.05%), its dominant performance in other key metrics (Rec, IoU, $F1$, Kappa, and OA) collectively proves that DHCNet effectively balances multidimensional evaluations and achieves excellent overall performance task on the CDD dataset.

DHCNet achieves the best performance on all three datasets, which is because the proposed method, in the encoding stage, uses a transformer-based encoder to capture global dependencies and explores spatio-temporal clues between multitemporal images through an interaction module. Then, it supplements detailed information via an

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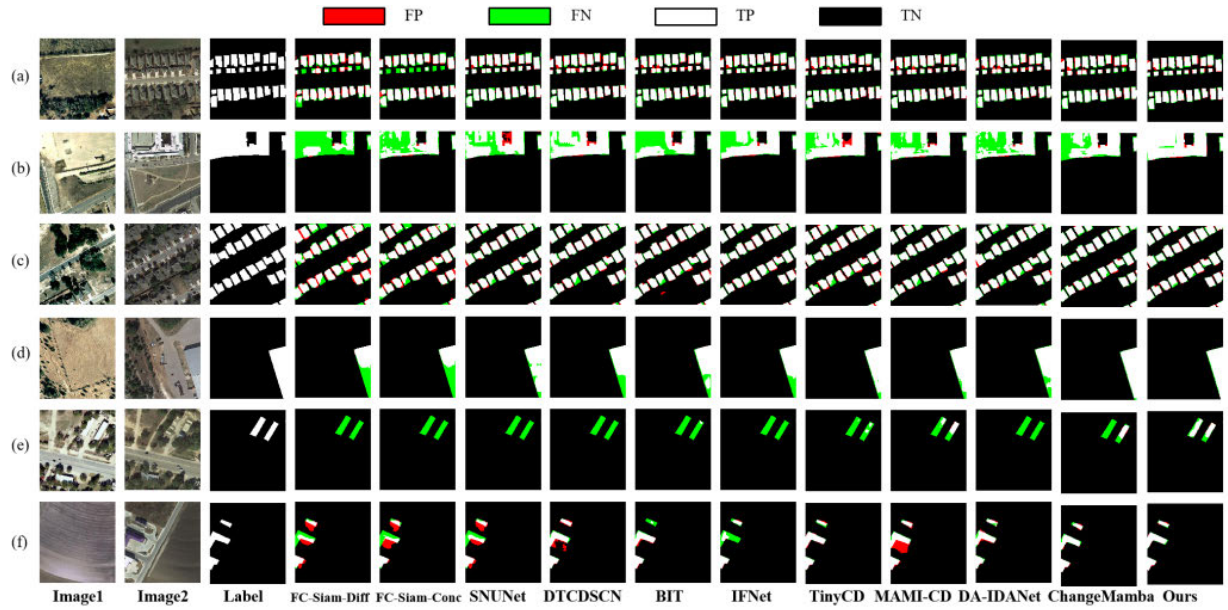


Fig. 6. Visual comparison of different methods on the LEVIR-CD dataset. Rows (a)–(f) correspond to representative test scenes with different change patterns and background complexities.

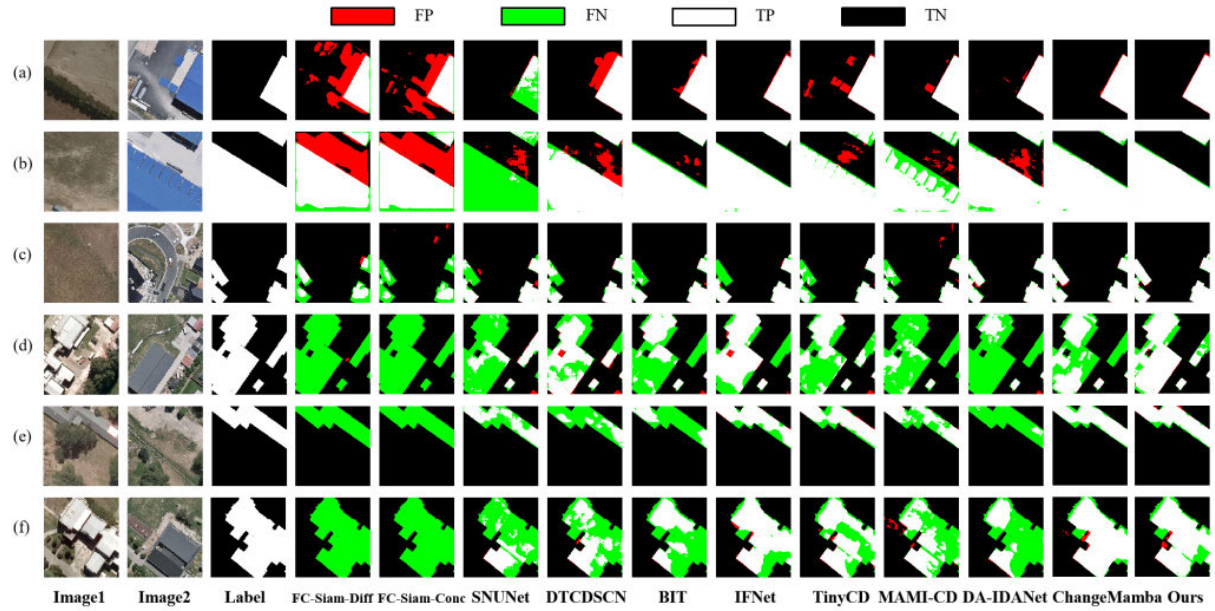


Fig. 7. Visual comparison of different methods on the WHU-CD dataset. Rows (a)–(f) correspond to representative test scenes with different change patterns and background complexities.

enhancement module. In the decoding stage, DHCNet fully utilizes multilevel features, integrating high-level semantic and spatial texture information from different levels into the final result. Thus, the performance of the model is enhanced.

D. Visual Results

The visualization results of each model on different datasets are shown in Figs. 6–8. These figures illustrate the accuracy of detecting change regions using different colors to represent TPs (white), TNs (black), FPs (red), and FNs (green). As shown in Fig. 6, six typical CD scenarios are presented from the LEVIR-CD dataset. In Fig. 6(a) and (c), DHCNet achieves more accurate predictions for dense, multiangle building changes. Its prediction results exhibit sharp boundaries with fewer missed detections (FN) and FPs, demonstrating the model's effectiveness in addressing issues such as blurred change boundaries, missed small objects, and misdetections. This effectiveness highlights DHCNet's capability to detect changes in complex building layouts. In Fig. 6(b) and (d), the compared methods are significantly inferior in detecting large buildings, with obvious large areas of incorrect predictions, further emphasizing the superiority of our approach.

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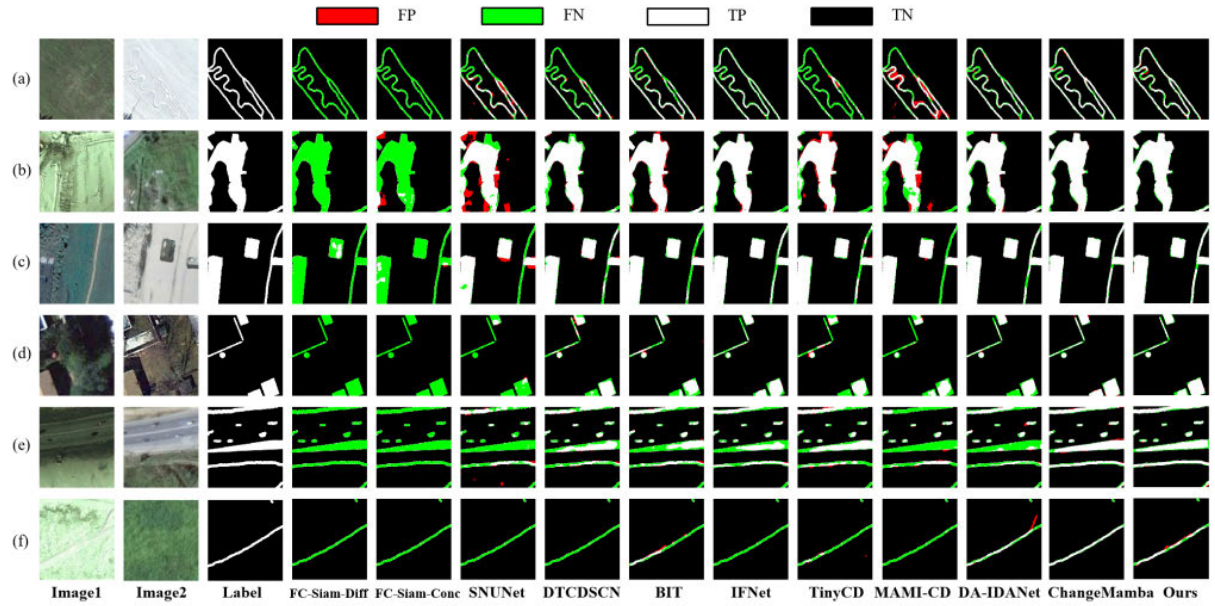


Fig. 8. Visual comparison of different methods on the CDD dataset. Rows (a)–(f) correspond to representative test scenes with different change patterns and background complexities.

The visualization results on the WHU-CD dataset are shown in Fig. 7. In Fig. 7(a) and (b), most compared methods (e.g., FC-Siam-Diff and IFNet) exhibit varying degrees of FPs. By contrast, DHCNet detects boundary changes more accurately, with its correctly identified regions closely aligning with actual change boundaries, demonstrating the method's fine-grained ability to capture building boundary alterations. For complex building cluster changes [see Fig. 7(d) and (f)], methods like FC-Siam-Conc and TinyCD show scattered misdetections or missed changes. In contrast, DHCNet's results closely match the ground truth, providing a more comprehensive detection of dense building changes. These results confirm the DHCNet's effectiveness in identifying fundamental changes within complex architectural layouts.

The visualization results on the CDD dataset are shown in Fig. 8. Fig. 8(a), (e), and (f) focuses on fine linear boundary changes, where methods such as TinyCD and MAMI-CD exhibit false detections that affect accuracy. In contrast, DHCNet's detection results closely align with the ground truth, showing almost no false positives, highlighting its precise ability to capture subtle boundary changes. For complex shape changes in Fig. 8(b), (c), and (e), methods like FC-Siam-Diff and SNUNet suffer from both false detections and missed changes. By comparison, DHCNet's correctly detected regions are complete, covering the main structural changes, indicating its more accurate discrimination when handling complex shape variations.

In summary, our findings reveal that CNN-based CD methods exhibit extensive incorrect predictions, whereas transformer-based approaches, capable of capturing global image information, outperform CNN-based counterparts. Beyond integrating these advantages, DHCNet further incorporates spatio-temporal correlations between multitemporal images, enabling superior detection performance in resisting pseudo-change interference, reducing missed and false detections. DHCNet maintains stable detection accuracy across dense building areas and mixed-type changes, reflecting its adaptability to diverse change scenarios. Notably, it demonstrates significant advantages in boundary prediction and complex change recognition, showcasing a more comprehensive perception of change regions and the ability to capture subtle change features.

TABLE V

ABLATION EXPERIMENTAL RESULTS OF DIFFERENT MODULES ON THE LEVIR-CD DATASET

DHCI	MSFE	MLFR	DS	F1(%)	IoU(%)	OA(%)	Kappa(%)
×	×	×	×	90.14	82.04	99.04	89.63
✓	×	×	×	91.19	83.69	99.13	90.66
✓	✓	×	×	91.31	84.00	99.14	90.85
×	✓	✓	✓	91.43	84.22	99.14	90.98
✓	×	✓	✓	91.39	84.14	99.15	90.94
✓	✓	×	✓	91.47	84.29	99.16	91.03
✓	✓	✓	×	91.33	84.04	99.15	90.88
✓	✓	✓	✓	91.55	84.42	99.16	91.12

TABLE VI

ABLATION EXPERIMENTAL RESULTS OF DIFFERENT MODULES ON THE WHU-CD DATASET

DHCI	MSFE	MLFR	DS	F1(%)	IoU(%)	OA(%)	Kappa(%)
×	×	×	×	93.01	86.93	99.41	92.70
✓	×	×	×	93.88	88.47	99.50	93.62
✓	✓	×	×	94.21	89.05	99.51	93.95
×	✓	✓	✓	94.71	89.95	99.56	94.48
✓	×	✓	✓	94.99	90.47	99.58	94.78
✓	✓	×	✓	94.59	89.74	99.55	94.35
✓	✓	✓	×	94.54	89.64	99.54	94.30
✓	✓	✓	✓	95.73	91.80	99.65	95.54

E. Ablation Analysis

This section conducted ablation analyses of DHCNet on three datasets. First, the effectiveness and rationality of each module were evaluated by removing the modules from the DHCNet. Second, comparative experiments were set up to examine the influence of the exchange ratio of the DHCI

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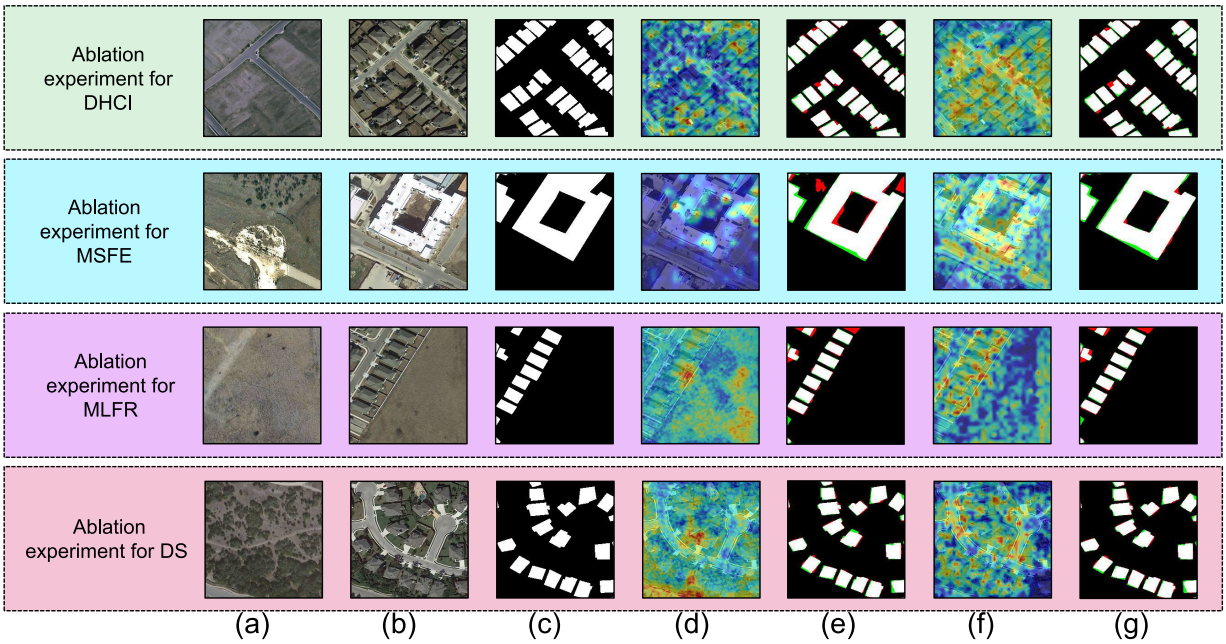


Fig. 9. Visual results of ablation experiments for removing different modules on the LEVIR-CD dataset. (a) T1 images. (b) T2 images. (c) Ground truth. (d) Feature visualization results of ablation experiments for different modules. (e) CD results of ablation experiments for different modules. (f) Feature visualization results of DHCNet. (g) CD results of DHCNet.

TABLE VII

ABLATION EXPERIMENTAL RESULTS OF DIFFERENT MODULES ON THE CDD DATASET

DHCI	MSFE	MLFR	DS	F1(%)	IoU(%)	OA(%)	Kappa(%)
×	×	×	×	94.82	90.15	98.81	94.15
✓	×	×	×	95.82	91.99	99.03	95.28
✓	✓	×	×	96.06	92.42	99.09	95.54
×	✓	✓	✓	96.15	92.58	99.10	95.64
✓	×	✓	✓	96.17	92.62	99.11	95.66
✓	✓	×	✓	96.14	92.56	99.10	95.63
✓	✓	✓	×	96.17	92.63	99.11	95.67
✓	✓	✓	✓	96.40	93.04	99.16	95.92

module and the weights of each branch in DS. Finally, we analyze the influence of feature interaction pattern, the resolution and bands of input images, and the weight ratio of the loss function on the accuracy of CD. The experimental results of different modules on the three datasets are presented in Tables V–VII, and the visual results is shown in Fig. 9.

1) DHCI Module: To evaluate the impact of the DHCI module on model performance, ablation experiments were conducted by removing the DHCI module from the network. When the DHCI module was excluded, the model's performance significantly declined across all three datasets: the $F1$-scores dropped to 94.71% (WHU-CD), 91.43% (LEVIR-CD), and 96.15% (CDD), respectively, while the IoU values decreased by 1.85%, 0.20%, and 0.46%, respectively. These experiments demonstrate that the DHCI module plays a critical role in enhancing feature interaction, improving discriminative feature extraction, and strengthening the perception of spatio-temporal cues, which are essential for accurate CD. The DHCI module focuses on the dynamic interaction of bi-temporal features and mainly consists of two core components: index-separable feature exchange and cross-temporal collaborative attention. The index-separable feature exchange can precisely control the interaction proportion, avoiding information redundancy caused by global feature exchange. It also preserves the independence of single features, where features in nonexchanged regions maintain their original temporal sequence. The cross-temporal collaborative attention not only focuses on features of a single-temporal phase but also explores complementary information between bi-temporal features through cross-temporal attention weight calculation. This complementary fusion enables the model to stably detect changes in complex scenarios and enhances the scene robustness of the method.

2) MSFE Module: After removing the MSFE module, the IoU values dropped to 90.47% (WHU-CD), 84.14% (LEVIR-CD), and 92.62% (CDD), respectively, and the Kappa coefficients decreased to 94.78%, 90.94%, and 95.66%, respectively. This performance decline is due to the model's lack of multiscale feature extraction ability, which weakens the ability to capture change features at different scales, leading to an overall decline in performance. The above experimental results validate the crucial role of the MSFE module in enhancing multiscale feature representation and enriching feature richness. The MSFE module comprises two major parts: differential feature extraction and feature enhancement. The differential feature extraction component captures the differences between bi-temporal images and fuses differential features with original features; the feature enhancement component captures and strengthens features of different scales, improves feature representation, and effectively enhances the accuracy and robustness of CD in complex scenarios.

3) MLFR Module: Ablation experiments removing the MLFR module were conducted to validate its impact on the detection accuracy. The Kappa coefficients dropped to 94.35%

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(WHU-CD), 91.03% (LEVIR-CD), and 95.63% (CDD) across the datasets, while IoU values decreased by 2.06%, 0.13%, and 0.48%, respectively. When the MLFR module is removed, the MLFR mechanism is absent, leading to ineffective recovery and fusion of feature information between levels. This absence weakens the feature representation ability, subsequently reducing the model's accuracy in identifying and localizing change features. The experimental results fully demonstrate that the MLFR module plays a critical role in maintaining the integrity of hierarchical feature information and improving the model's detection accuracy for change features. The MLFR module achieves MLFR and fusion through the collaboration of upper and lower components: the upper component extracts global statistical features, generates attention weights to weight the input features, and strengthens key features; the lower component performs feature enhancement by generating attention weights and then extracts contextual information. Finally, the features from the previous decoder stage are concatenated with those processed by the two components for multiscale feature fusion, which effectively improves the semantic consistency and detail recovery capability of the features, thereby enhancing the CD accuracy.

To explore the interaction effects between modules, the ablation experimental results show that the DHCI+MSFE combination significantly outperforms the model with only the DHCI module across all core metrics on all datasets, while it is outperformed by the full DHCI+MSFE+MLFR combination. These results fully demonstrate the existence of a significant synergistic effect between the DHCI module's capability of dynamic bi-temporal feature interaction and the MSFE module's functions of differential feature extraction and multiscale enhancement. Specifically, the refined interactive features output by the DHCI module provide a high-quality input foundation for the MSFE module's difference capture; the further feature enhancement by the MSFE module better supports subsequent CD tasks; and the MLFR module performs recovery and reconstruction of multilevel features. Their combination effectively improves the overall performance of the model.

4) Deep Supervision: Ablation experiments were conducted by removing the DS module. The $F1$-scores of the DHCNet model dropped to 94.54% (WHU-CD), 91.33% (LEVIR-CD), and 96.17% (CDD), respectively. The DS module embeds supervision signals at different network layers to provide multistage guidance during model training. This mechanism optimizes the gradient backpropagation path, mitigates the gradient vanishing issue in deep network training, and promotes the learning of more discriminative feature representations at each network layer.

The DS mechanism of DHCNet consists of a main output and multiple auxiliary outputs. The main output is responsible for the final high-Pre CD results, while the auxiliary outputs serve to provide gradient guidance during training and enhance shallow-level detail information. The weights of auxiliary outputs must be lower than that of the main output to avoid interfering with the core prediction. In the optimization process, first, all auxiliary branches are set to equal weights to verify the basic gradient guidance effect of the auxiliary outputs. Subsequently, considering that the auxiliary branch closer to the main output has larger feature sizes and richer spatial details, offers higher auxiliary value for the final prediction. Thus, the weight of the branch is adjusted to be higher than those of other shallow auxiliary branches. Finally, on the basis of hierarchical adaptation, the overall weights of the auxiliary branches are gradually increased and validated on different datasets. It is found that this adjustment enables the model to achieve rapid convergence in the early training stage and ultimately obtains optimal performance on all datasets, leading to the determination of the final weight combination.

In Table VIII, the DS branch weight combination [0.5, 0.5, 0.8, 1.0] demonstrates the best generalization capability. On the CDD dataset, this configuration enables DHCNet to achieve $F1$/IOU scores of 96.40%/93.04%, representing improvements of 0.36% in $F1$ and 0.66% in IOU compared to the suboptimal weight combination [0.4, 0.4, 0.6, 1.0]. On other datasets, the DS branch weight combination [0.5, 0.5,

TABLE VIII

$F1$/IoU OF DHCNET WITH VARIOUS DS RATES

DS rate	LEVIR-CD		WHU-CD		CDD	
	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)
0.5, 0.5,0.5, 1.0	91.34	84.33	94.95	90.39	96.18	92.65
0.4, 0.4,0.6, 1.0	91.35	84.08	95.56	91.49	96.04	92.38
0.5, 0.5,0.8, 1.0	91.55	84.42	95.73	91.80	96.40	93.04

TABLE IX

$F1$/IoU OF DHCNET WITH VARIOUS FEATURE EXCHANGE RATIOS

Rate P	LEVIR-CD		WHU-CD		CDD	
	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)
Learnable parameter	91.64	84.58	95.62	91.62	95.83	92.00
0	91.35	84.09	95.21	90.86	95.80	91.95
1/8	91.45	84.24	95.20	90.84	96.31	92.89
1/4	91.55	84.42	95.73	91.80	96.40	93.04
1/2	91.33	84.04	95.56	91.54	96.22	92.72

TABLE X

$F1$/IoU OF DHCNET WITH VARIOUS EXCHANGE MODE

Mode	LEVIR-CD		WHU-CD		CDD	
	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)
C	91.23	83.87	95.30	91.02	95.74	91.83
S	91.31	84.01	95.32	91.06	95.83	91.99
C-S	91.26	83.92	95.15	90.95	95.75	91.85
S-C	91.55	84.42	95.73	91.80	96.40	93.04

TABLE XI

$F1$/IoU OF DHCNET WITH VARIOUS RESOLUTIONS OF INPUT IMAGES

Resolution	LEVIR-CD		WHU-CD		CDD	
	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)
64 × 64	85.66	74.91	92.91	86.76	89.36	80.77
128 × 128	90.20	82.14	95.16	90.77	93.41	87.63
256 × 256	91.55	84.42	95.73	91.80	96.40	93.04

TABLE XII

$F1$/IoU OF DHCNET WITH VARIOUS LOSS FUNCTION WEIGHT SETTINGS

Weight	LEVIR-CD		WHU-CD		CDD	
	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)
1:1	91.55	84.42	95.73	91.80	96.40	93.04
1:2	90.56	82.76	94.75	90.02	92.97	86.86
2:1	90.94	83.39	94.87	90.26	94.03	88.74

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TABLE XIII
*F*1/IoU OF DHCNET WITH VARIOUS BANDS OF INPUT IMAGES

Band	LEVIR-CD		WHU-CD		CDD	
	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)
R	89.82	81.52	94.48	89.54	92.14	85.42
R+G	91.08	83.62	95.11	90.69	95.68	91.72
R+G+B	91.55	84.42	95.73	91.80	96.40	93.04

0.8, 1.0] also maintains its advantage: the $F1$-score on the WHU-CD dataset reaches 95.73%, leading other configurations by 0.17%, and the IoU on the LEVIR-CD dataset reaches 84.42%, outperforming other configurations by 0.09%. These results indicate that reinforcing terminal supervision weights balances the contribution of features at different levels and effectively guides the learning direction of multiscale features.

Fig. 9(e), which excludes different modules, exhibits significant misdetections and missed detections compared to Fig. 9(g), with all modules included. In addition, the attention map in Fig. 9(d) shows scattered focus, failing to concentrate on critical changed regions, unlike the focused attention in Fig. 9(f). This visual evidence further validates the necessity of the four modules in improving detection accuracy and focusing on key change areas.

5) Feature Interaction Pattern in DHCI Module: Parameter P is the core hyperparameter that controls the information exchange ratio of bi-temporal features in the spatial or channel dimension; therefore, the value of P needs to be verified through experiments. Comparative experiments were conducted by setting $P = 0$, $P = 1/8$, $P = 1/4$, $P = 1/2$, and P as a learnable parameter, with the experimental results shown in Table IX. When P is set to 1/4, the DHCI module achieves peak performance on the CDD and WHU-CD datasets, with its $F1$-scores reaching 96.40% and 95.73%, respectively. On the LEVIR-CD dataset, the $F1$-score and IoU of the DHCI module reach 91.55% and 84.42%, respectively. When $P = 1/8$ or $P = 1/2$, the experimental results are worse than those when $P = 1/4$. This result indicates that either too high or too low an exchange ratio will affect the performance of the proposed method. When $P = 0$, no feature exchange is performed, and the experimental results are worse than those of other experiments. This result demonstrates that the proposed method is sensitive to P. In addition, we investigated the impact on model performance by setting P as a learnable parameter. The results show that on the LEVIR-CD dataset, the performance of this setting is slightly better than that of the fixed exchange ratio of 1/4; however, on the CDD and WHU-CD datasets, its performance is inferior to that of the exchange ratio of 1/4. These results indicate that the exchange ratio 1/4 has stronger stability and advantages, showing better parameter robustness.

Through feature exchange, the contextual information of bi-temporal features can be perceived by means of mutual learning. On the other hand, the feature distributions of the two branches become more similar, automatically achieving a certain degree of domain adaptation between the bi-temporal features [52]. The DHCI module combines spatial/channel exchange based on index divisibility, and it has the following main advantages: First, feature exchange only covers positions/channels divisible by "$1/P$," but the attention mechanism also indirectly enables nonexchanged positions/channels to learn the correlations between bi-temporal features through weight propagation. Second, regularized exchange is single-dimensional, while the attention mechanism can learn cross-position-channel composite correlations, covering more complex change patterns. Different heads of the multihead attention mechanism can focus on different scales, which complements the "fine/coarse granularity" of feature exchange—this allows the model to grasp detailed and global changes simultaneously. Therefore, it is feasible to introduce the combination of spatial/channel exchange based on index divisibility and the transformer architecture into the DHCNet.

To further verify the sensitivity of the exchange strategy (i.e., spatial/channel exchange based on index divisibility), we designed four schemes to compare different feature exchange strategies: channel-only exchange (C) strategy, spatial-only exchange (S) strategy, channel-then-space exchange (C-S) strategy, and space-then-channel exchange (S-C) strategy. As shown in Table X, the S-C strategy outperforms the other three schemes in most metrics across the three datasets: These results indicate that performing spatial exchange first and then channel exchange enables more efficient feature interaction, which helps capture complex change patterns in images and improves CD performance.

6) Resolution of Input Images: To verify the impact of different spatial resolutions on CD accuracy, we conducted ablation experiments by downsampling the original 256×256 images to spatial resolutions of 64×64 and 128×128, followed by retraining the model parameters and evaluating the performance. The experimental results are shown in Table XI. When the resolution decreases from 256×256 to 128×128 and then to 64×64, the model's CD performance decreases gradually. Taking the LEVIR-CD dataset as an example: at 256×256, $F1$ is 91.55%, and IoU is 84.42%; at 128×128, $F1$ drops to 90.20% and IoU to 82.14%; at 64×64, $F1$ further decreases to 85.66% and IoU to 74.91%. The WHU-CD and CDD datasets show the same trend: the lower the resolution, the lower the $F1$ and IoU. High-resolution inputs preserve richer spatial details (for example, edges and fine-grained changes), enabling more accurate change region localization and segmentation.

7) Loss Function Weight Configuration: To investigate the impact of balancing BCE loss and Dice loss on CD performance, we conducted comparative experiments with different loss function weight configurations. The results are shown in Table XII. On three different datasets, the 1:1 weight configuration achieves the highest $F1$ and IoU scores, outperforming other weight configurations. These results indicate that equal weighting of BCE loss and Dice loss better balances the CD accuracy and change region integrity. Over-emphasizing either loss impairs performance and weakens the ability to optimize pixel-level classification and region-level change integrity synergistically.

8) Bands of the Input Images: To analyze the impact of spectral band differences on CD accuracy, this article conducts

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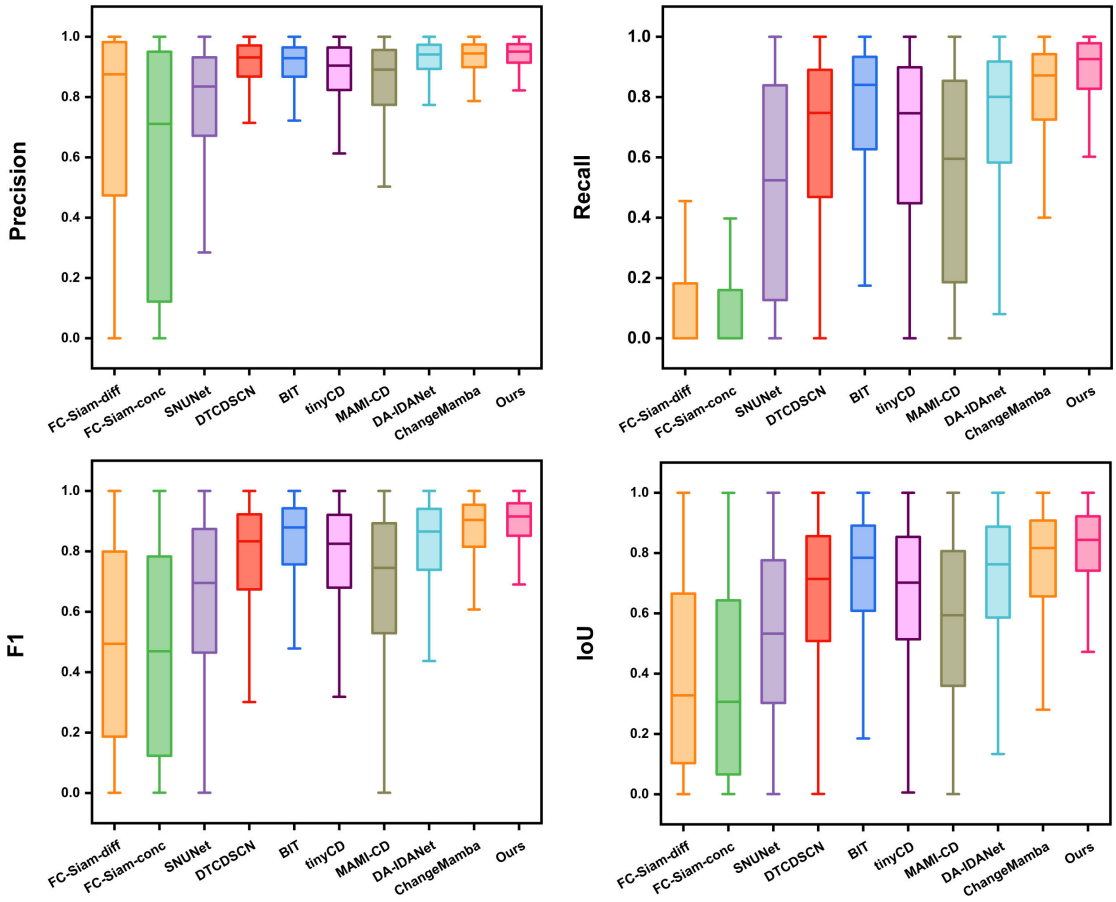


Fig. 10. Comparison of Pre, Rec, $F1$, and IoU for CD results across different methods on the CDD dataset.

ablation experiments on different band combinations [R (single band), RG (two bands), RGB (three bands)]. The results are shown in Table XIII. Increasing the number of bands improves $F1$ and IoU scores on all three datasets. Additional bands provide richer spectral information, helping the model distinguish different ground objects, capture subtle spectral differences, and thus improve the accuracy of change region localization and segmentation.

TABLE XIV
EFFICIENCY ANALYSIS RESULTS OF VARIOUS METHODS

Method	Parameter(M)	FLOPs(G)	FLOPs per Parameter(G/M)	GPU-Inference(s)	GPU-Training-epoch(s)	GPU-Train(G)	GPU-Inference(G)	CPU-Inference(s)
FC-EF	1.35	3.13	2.32	0.011	40	3.53	1.30	0.024
FC-Siam-diff	1.35	4.29	3.17	0.020	45	4.51	1.35	0.028
FC-Siam-conc	1.55	4.89	3.15	0.020	45	4.63	1.35	0.028
SNUNet	12.04	46.82	3.89	0.017	140	14.82	5.25	0.271
IFNet	36.00	79.04	2.20	0.018	144	12.22	5.35	0.273
DTCDSCN	41.07	11.51	0.28	0.022	116	4.51	1.44	0.072
BIT	11.91	8.53	0.72	0.019	50	6.52	0.98	0.054
ChangeFormer	41.03	139.00	3.39	0.021	194	19.26	6.71	0.357
tinyCD	0.29	1.54	5.31	0.019	60	10.35	1.66	0.061
MAMI-CD	2.02	17.88	8.85	0.013	80	11.69	1.94	0.128
DA-IDANet	17.11	49.15	2.87	0.017	191	19.96	6.78	0.187
ChangeMamba	49.94	114.82	2.30	0.017	270	17.26	4.95	-
Ours	42.93	21.52	0.50	0.020	142	16.51	5.52	0.199

F. Analysis of Model Accuracy and Complexity

To obtain the data distribution of quantitative metrics, we conducted a more in-depth analysis on the CDD dataset. Fig. 10 presents a comparative analysis of boxplots for four metrics—Pre, Rec, $F1$, and IoU—across different methods on the CDD dataset.

The $F1$, IoU, Rec, and Pre metrics were calculated for each test image, and their boxplots were generated. In the Pre metric, the median value of the boxplot for the proposed method, significantly higher than those of other methods. The box of the DHCNet is highly concentrated with a smaller fluctuation range, indicating superior Pre in determining change regions and stronger stability. In the Rec metric, the boxplot of the proposed method also shows a high position, reflecting a higher ability to capture real change regions. Further analysis of the $F1$ and IoU metrics reveals that the median values of the boxplots for the proposed DHCNet significantly exceed those of methods such as SNUNet and DTCDSCN, with smaller fluctuation ranges. These analysis results validate its excellent ability to balance Pre and Rec and highlight the advantage in the accuracy of change region localization. In summary, DHCNet demonstrates superior performance and stability across all four core metrics.

This study presents statistics on the parameter, FLOPs, FLOPs per parameter, per-sample inference time, per-epoch training time under a GPU environment, GPU memory usage during training, GPU memory usage during inference, and per-sample inference time under a CPU (i5-14600KF) environment of different methods. Note that during statistics, the input data size was set to $3 \times 256 \times 256$, with data sourced from the LEVIR-CD dataset. The results are shown in Table XIV. The proposed DHCNet has a parameter size of 42.93 M, primarily composed of 33.69 M from the Swin transformer backbone and channel adaptation, 3.92 M from

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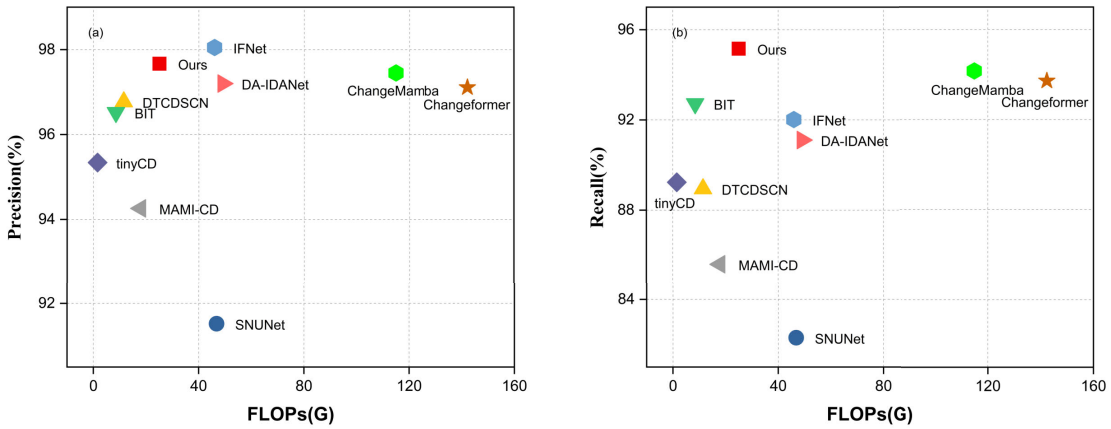


Fig. 11. Relationship between model's FLOPs and its (a) Pre and (b) Rec performance on CDD dataset.

DHCI modules, 2.77 M from MSFE modules, 1.58 M from MLFR modules, and 0.40 M from the classifiers. DHCNet maintains the highest accuracy while having a parameter size only slightly larger than that of peer methods. For FLOPs, 15.92 G comes from the Swin transformer backbone and channel adaptation, 2.86 G from DHCI modules, 1.18 G from MSFE modules, 0.60 G from MLFR modules, and 0.88 G from the classifiers, which is significantly lower than that of many advanced comparative methods such as SNUNet, IFNet, and ChangeFormer. In addition, DHCNet achieves a moderate level in terms of inference time under both CPU and GPU environments, training time, as well as GPU memory usage during the training and inference phases. This is attributed to the architectural design of our DHCNet, particularly the efficiency of feature interaction and simplified module design, which reduces redundant computations without sacrificing performance. Future work will achieve lightweight methods, further improving the applicability of DHCNet in practical CD scenarios.

We also analyzed the relationship between the model's FLOPs and Pre/Rec metrics on the CDD dataset, as shown in Fig. 11. In Fig. 11(a), which examines the correlation between Pre and FLOPs, DHCNet achieves the second-highest Pre value with relatively low FLOPs, significantly outperforming methods such as SNUNet and MAMI-CD. Fig. 11(b) further shows that DHCNet leads in the Rec metric without incurring excessively high FLOPs. Although Changeformer exhibits prominent Rec performance, its FLOPs are far higher than that of DHCNet. These results confirm that DHCNet maintains high Pre and Rec while achieving leading $F1$-scores among comparative methods. By effectively controlling computational load and avoiding excessive computational costs, DHCNet demonstrates excellent synergy between model efficiency and performance.

In CD tasks, the tradeoff between computational complexity and detection performance is a key issue that needs to be emphasized in practical applications. Over-pursuing high performance while ignoring complexity leads to a sharp increase in model deployment costs; conversely, one-sidedly reducing complexity may sacrifice detection accuracy, failing to meet the requirements of fine-grained CD. DHCNet achieves an excellent balance through a modular architecture design. In terms of computational complexity, it is significantly lower than methods such as SNUNet, ChangeFormer, and ChangeMamba; meanwhile, its inference time, training time, and GPU memory usage are all at moderate levels. In terms of performance, the CD results of DHCNet outperform all comparative methods, significantly reducing redundant computations while ensuring performance. Most existing models have obvious shortcomings in balancing complexity and performance. For example, although ChangeFormer performs well in the Rec metric, its FLOPs are excessively high, leading to a significant increase in deployment costs; lightweight models such as tinyCD, while having extremely low complexity, are significantly inferior to mainstream methods in performance. In contrast, DHCNet's advantage lies in achieving comprehensive performance leadership with moderate computational complexity, especially in the localization accuracy of changed regions, demonstrating tremendous advantages in practical scenarios.

For practical applications, the criterion for judging additional computational costs is that the newly added computational load is at a slight level while bringing significant accuracy improvement. Ablation experiments show that in accuracy-priority scenarios for CD model deployment, minor computational overhead is well worth the cost for substantial accuracy gains. For example, adding the DHCI module only increases the model's FLOPs by 15.32% and the parameter count by 10.05%, but improves the $F1$-score by 1.05% and IoU by 1.65% on the LEVIR-CD dataset, effectively reducing the missed detection of small changed regions.

G. Comparative Analysis of Challenging Regions

To further verify the proposed method's performance in various challenging scenarios, we conducted visual comparisons on complex cases (e.g., blurred boundaries, subtle changes, registration imbalance, and solar altitude angle differences) from the CDD dataset. The results are shown in Fig. 12. In the blurred boundary scenario, pixel features transition indistinctly at the boundaries of ground objects. Other methods exhibit obvious missing detection regions and fail to capture changes

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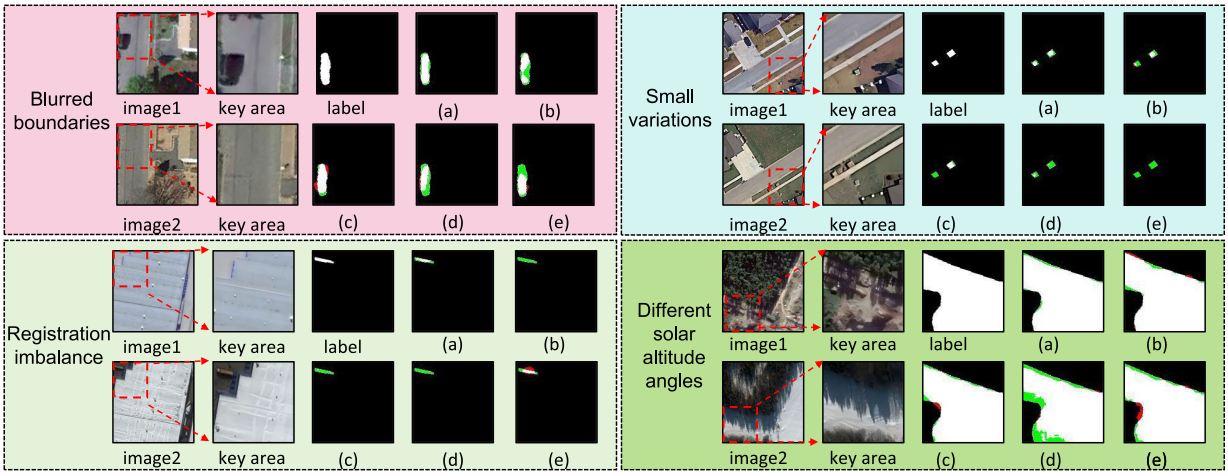


Fig. 12. Comparison of challenging regions on the CDD dataset. (a) DHCNet. (b) BIT. (c) DA-IDANet. (d) MAMI-CD. (e) tinyCD. "Key area" represents the magnified views of challenging regions. Label represents the ground truth of the key area.

TABLE XV
QUANTITATIVE COMPARISON RESULTS IN CHALLENGING SCENARIOS

Methods	Blurred boundaries		Small variations		Registration imbalance		Different solar altitude angles	
	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)	IoU	F1(%)	IoU(%)
tinyCD	67.16	50.55	0.00	0.00	56.98	39.84	97.33	94.80
MAMI-CD	67.52	50.97	39.25	24.42	1.05	0.53	93.30	87.45
DA-IDANET	74.92	63.79	59.63	42.48	5.15	2.65	98.50	97.04
BIT	60.61	43.49	74.21	59.00	17.86	9.80	98.27	96.61
DHCNet	80.94	67.98	84.28	72.83	61.03	43.92	99.51	99.02

at blurred boundaries; In contrast, the proposed method can reduce the gap between the detection results and the ground truth, but it also has the issue of imprecise boundaries of the change regions. In the small variation scenario, other methods have obvious missing detections, and cannot accurately identify such subtle changes. DHCNet fully extracts multiscale features corresponding to subtle changes and integrates semantic and texture information from features of different levels into the final detection results, thereby improving model performance.

In the registration imbalance scenario, specific registration deviations between the bi-temporal images lead to offsets in the positional of ground objects. Comparative methods generally suffer from missing or false detections; the proposed method, by means of DHCI, compensates for feature registration caused by misalignment imbalance and can still relatively accurately detect real change regions. In the different solar altitude angle scenario, images under different sun elevation angles have different lighting conditions. Other methods have many missing detections, while the proposed DHCNet leverages the synergy of DHCI, MSFE, and MLFR to distinguish between changes caused by lighting differences and real changes.

We further compare the $F1$-scores and IoU of DHCNet with existing mainstream methods in the challenging scenario shown in Fig. 12, and the quantitative experimental results are shown in Table XV. In the blurred boundaries scenario, the performance of all methods degrades to a certain extent, but DHCNet's $F1$ (80.94%) and IoU (67.98%) are significantly superior to those of other comparative methods, with the smallest performance degradation. In the small variations scenario, some methods (e.g., tinyCD) can barely capture subtle changes, while DHCNet's $F1$ (84.28%) and IoU (72.83%) are far ahead of other methods, demonstrating a stronger ability to capture subtle changes. In the registration imbalance scenario—which exhibits the most significant performance degradation—DHCNet's $F1$ (61.03%) and IoU (43.92%) are much higher than those of other comparative methods, reflecting a stronger ability to resist registration errors. In the different solar altitude angles scenario, the performance of all methods is relatively stable, and DHCNet's $F1$ (99.51%) and IoU (99.02%) still maintain the optimal level. Although the model performance degrades to a certain extent in most challenging scenarios, DHCNet outperforms all existing comparative methods, verifying its robustness in practical complex tasks.

For the analysis of extremely small targets and fine-grained changes, this article calculates the proportion of changed pixels under different scenarios. The registration imbalance scenario corresponds to a changed proportion of 0.62%; the blurred boundary scenario has a changed proportion of 3.68%; and the small variations scenario has a changed proportion of only 0.5%. Combined with the quantitative experimental results, in the extremely fine-grained scenario, DHCNet still outperforms existing methods, fully verifying its effective ability to capture extremely small targets and fine-grained changes.

VI. MODEL GENERALIZATION ABILITY TESTING AND LIMITATION ANALYSIS

A. Comparative Experiments and Analysis in Real-World Scenario

To further verify the effectiveness and superiority of the proposed method in real HRRS images, we conducted CD experiments on the Real-Wuhan dataset. Table XVI presents the quantitative results of different comparison methods on the Real-Wuhan dataset.

The proposed DHCNet shows significant advantages on various evaluation metrics. In terms of Rec, DHCNet reaches 76.66%, which is far ahead of other models. This means it can more comprehensively capture the real existing change regions and effectively reduce missed detections. The IoU

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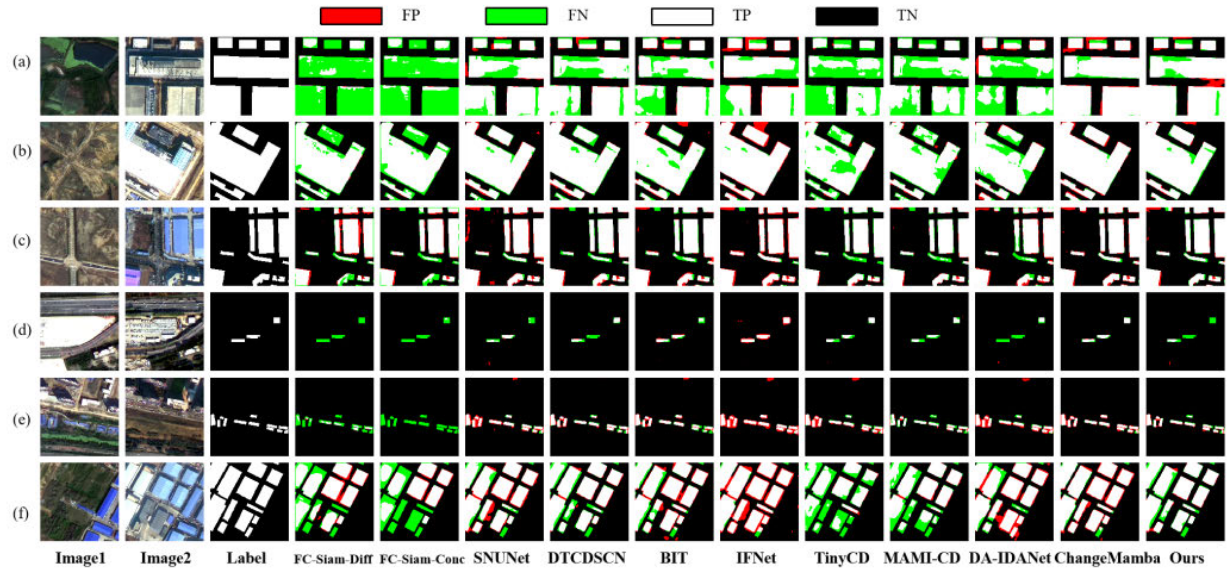


Fig. 13. Visual comparison of different methods on the Real-Wuhan dataset. Rows (a)–(f) correspond to representative real-world remote sensing test scenes with different change patterns and background complexities.

TABLE XVI
QUANTITATIVE COMPARISON RESULTS ON THE REAL-WUHAN DATASET

Method	Pre(%)	Rec(%)	IoU(%)	F1(%)	Kappa(%)	OA(%)
FC-EF	**84.49**	17.39	16.86	28.85	27.69	95.99
FC-Siam-diff	83.23	29.79	28.10	43.88	42.46	96.44
FC-Siam-conc	87.07	27.71	26.62	42.04	40.71	96.43
SNUNet	72.57	65.57	52.55	68.89	67.45	97.23
IFNet	71.84	73.66	57.16	72.74	71.38	97.42
DTCDSCN	78.09	63.68	54.03	70.15	68.85	97.47
BIT	77.30	63.81	53.74	69.91	68.58	91.43
ChangeFormer	76.91	63.34	53.22	69.47	68.12	97.40
tinyCD	77.51	65.32	54.91	70.89	69.59	97.49
MAMI-CD	77.79	60.52	51.60	68.07	66.71	97.35
DA-IDANet	74.28	61.91	50.98	67.54	66.09	97.22
ChangeMamba	82.18	63.70	55.98	71.77	70.57	97.66
Ours	74.63	**76.66**	**60.81**	**75.63**	**74.42**	**97.69**

of the DHCNet is 60.81%, which is significantly improved compared with better-performing comparison models such as IFNet (57.16%) and tinyCD (54.91%), reflecting a higher overlap between predicted change regions and real change regions. The $F1$-score of the DHCNet is 75.63%, which has outstanding advantages in comprehensively balancing Pre and Rec. In contrast, other methods struggle to effectively meet complex CD requirements in real-world scenarios, due to their limited feature extraction and fusion capabilities. The proposed method, however, benefits from the efficient capture of cross-temporal feature dependencies. DHCNet also achieves full exploitation of multiscale texture and semantic information. This enables the DHCNet to have stronger perception and detection capabilities for diverse and complex changes in real datasets.

To more intuitively demonstrate the performance of the proposed method on the Real-Wuhan dataset, visual results are shown in Fig. 13. In Fig. 13(b), methods such as SNUNet and DTCDSCN have edge false detection and internal missing detection when detecting newly built buildings. This problem causes the change regions to have blurred edges and incomplete interiors. In contrast, the proposed method clearly presents the entire change morphology of the buildings. In Fig. 13(c), when other methods handle changes of ground objects with different scales and irregular shapes, they have many false and missing detections. This phenomenon easily leads to inaccurate segmentation of change regions. The proposed DHCNet accurately distinguishes the change boundaries of different ground objects by effective fusion of multiscale features. For subtle changes in Fig. 13(d) and (e), most comparison methods have a large number of missing detection regions and fail to capture these subtle changes. However, by its ability to perceive fine-grained features, the proposed DHCNet successfully covers these subtle changes with a few missing detection regions. In the complex scene of dense buildings [see Fig. 13(f)], the proposed DHCNet can accurately identify changes of various ground objects. In contrast, other models have more false or missing detection regions and cannot fully detect the change regions.

B. Comparative Experiments and Analysis on SAR Images

To comprehensively evaluate the proposed method's performance on the SAR dataset, we compared it quantitatively with several mainstream CD models. Results are in Table XVII.

Regarding the IoU metric, the proposed DHCNet achieves 92.98%, significantly outperforming other comparative methods. Regarding the $F1$-score, the proposed DHCNet also achieves an excellent performance of 96.36%. It has an outstanding advantage in balancing Pre and Rec, demonstrating excellent ability to capture change regions. Other methods struggle to meet the CD requirements for SAR data accurately. Due to the lack of efficient capture of cross-temporal feature dependencies, these methods cannot perceive complex changes in SAR images.

To more intuitively demonstrate the performance of the proposed method on the S1GFloods dataset, we visually compared

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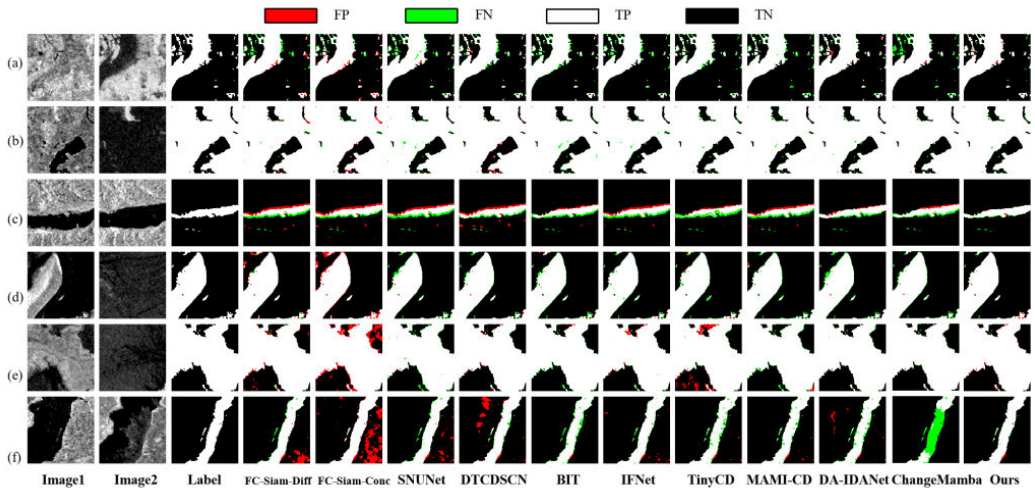


Fig. 14. Visual comparison of different methods on the S1GFloods dataset. Rows (a)–(f) correspond to representative SAR test scenes with different change patterns and background complexities.

TABLE XVII
QUANTITATIVE COMPARISON RESULTS ON THE S1GFLOODS DATASET

Method	Pre(%)	Rec(%)	IoU(%)	F1(%)	Kappa(%)	OA(%)
FC-EF	90.70	91.17	83.37	90.93	86.45	94.00
FC-Siam-diff	91.06	91.75	84.17	91.41	87.15	94.31
FC-Siam-conc	87.95	95.27	84.27	91.47	87.01	94.13
SNUNet	94.39	82.30	76.46	93.30	84.98	97.01
IFNet	95.19	92.24	87.45	94.33	90.06	95.63
DTCDSCN	91.07	95.96	87.71	93.45	90.10	95.56
BIT	94.57	93.93	89.12	94.25	91.43	96.22
ChangeFormer	95.98	93.66	90.12	94.80	92.29	96.61
tinyCD	94.22	95.61	90.31	94.91	92.37	96.61
MAMI-CD	95.45	93.83	89.83	94.64	92.04	96.49
DA-IDANet	93.53	95.43	89.52	94.47	91.71	96.31
ChangeMamba	95.71	94.01	90.22	94.86	92.36	96.64
Ours	**96.13**	**96.60**	**92.98**	**96.36**	**94.56**	**97.59**

the proposed method and several mainstream CD methods in Fig. 14. In the complex postflood change case [see Fig. 14(a)], compared with the proposed DHCNet, the remaining methods all have more severe missing detection issues. Comparative methods fail to accurately capture changes in low-contrast regions and changes of objects in small-scale areas, leading to significant missing detection regions. For complex edge change regions [see Fig. 14(c)], especially in river edges, comparative methods have more false and missing detections. These methods fail to fully utilize the spatial information of images when processing complex edge changes, leading to imprecise segmentation. By comparison, the proposed DHCNet can effectively capture the linear features of change regions through multiscale feature fusion. In Fig. 14(e), when handling large-scale irregularly shaped change regions, the proposed DHCNet achieves in-depth perception of SAR image features through the DHCI, enabling accurate detection of change regions.

C. Model Limitations and Future Work

Although the proposed DHCNet can effectively capture change information accurately in HRRS images, it adopts a relatively deep feature extraction backbone and complex interaction modules, which results in a high parameter count. Such an issue makes it difficult to deploy the DHCNet in low-computing-power scenarios, such as real-time UAV monitoring and on-orbit rapid processing of satellite data. In future research, we plan to design a lightweight backbone suitable for CD tasks and optimize the model using knowledge distillation technology to enrich the application scenarios of the DHCNet.

In addition, DHCNet currently can only detect changes in same-modal bi-temporal images, such as bi-temporal optical or SAR images. DHCNet does not yet support CD tasks for cross-modal bi-temporal images, such as the combination of a SAR image with a multispectral image, or point cloud data with an optical image. We plan to investigate cross-modal contrastive learning to enhance DHCNet's ability to detect changes in ground objects across different modalities.

VII. CONCLUSION

This study proposes a dynamic hierarchical collaborative learning method named DHCNet for HRRS CD tasks. DHCNet integrates the DHCI module to model the spatio-temporal dependencies between bi-temporal images, effectively mitigating the domain discrepancy issue of bi-temporal features. Meanwhile, the MSFE module is designed to extract and enhance texture and semantic information at different scales, improving the detection capability for small-scale and boundary changes. In addition, the MLFR module is proposed to fuse hierarchical features, reconstruct critical change information, and realize the accurate location of change boundaries. Finally, a DS mechanism is introduced to optimize feature learning at different decoding layers, ensuring training stability and generalization ability.

Experimental results conducted on the LEVIR-CD, WHU-CD, CDD, Real-Wuhan, and S1GFloods datasets have demonstrated that DHCNet outperforms state-of-the-art methods, achieving more accurate detection results in both public and real-world CD tasks. Ablation experiments have verified the necessity of each module in DHCNet and explored the impact of loss function weight configuration, the resolution, and the bands of input images on the CD accuracy. The proposed DHCNet can facilitate the rapid monitoring of scenarios such as floods and urban expansion, providing efficient

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technical support for disaster prevention and mitigation as well as resource management. Future work will focus on designing lightweight architectures to enhance model deployment efficiency and exploring cross-modal generalization techniques to support multisource remote sensing data, thereby increasing the applicability of DHCNet in various application scenarios.

ETHICAL USE

The method proposed in this study may present potential dual-use risks. The authors suggest that users comply with relevant laws, regulations, and data agreements, and apply the method appropriately for legitimate research and applications to avoid misuse.

DECLARATION OF INTERESTS

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

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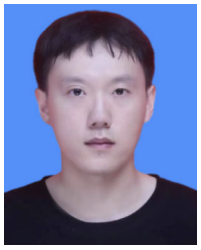
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